

UNIT-1. Part-A

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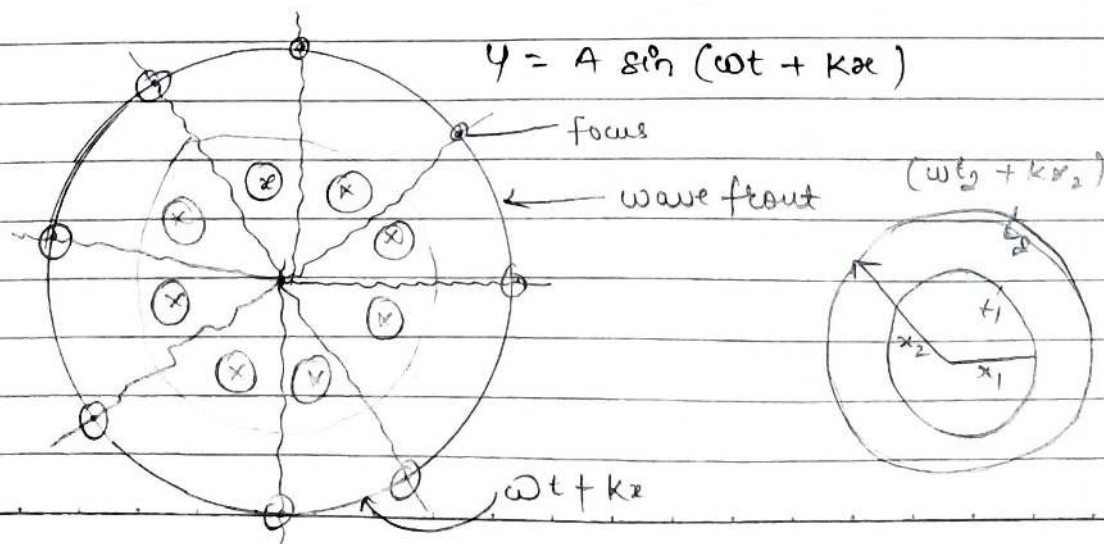
WAVE OPTICS

* Newton's Corpuscular Theory :

- Light is made with tiny particles.
- Every source of light emits particles called corpuscular.
- Corpuscular has different shape and size.
- Corpuscular are massless, elastic, and rigid in nature.
- Speed of corpuscular is more in denser medium than in rarer medium. (X)
- * → Later on 'Focault' has practically prove that speed of light is more in air than in water medium.
- * → So, this theory failed, but explained some phenomena like Rectilinear propagation, reflection & refraction of light.

* Huygen's Principle wave theory :

- ①. wave front :- "It is the locus of all points of a medium vibrating in a same phase".



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② Direction of propagation of wave is always perpendicular to the wave front.

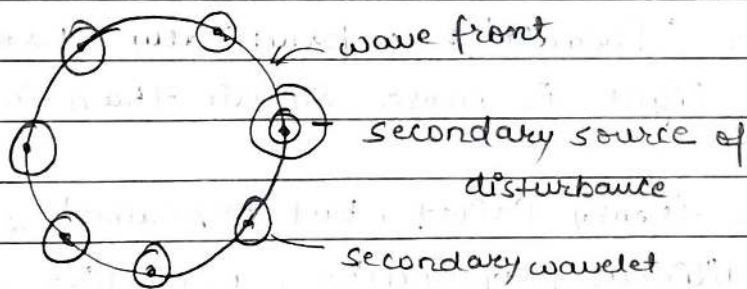
(a). for point source — sphere.

(b). for line source — cylinder.

(c). for light coming from ∞ — Plane.

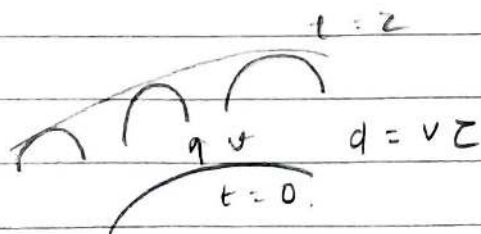
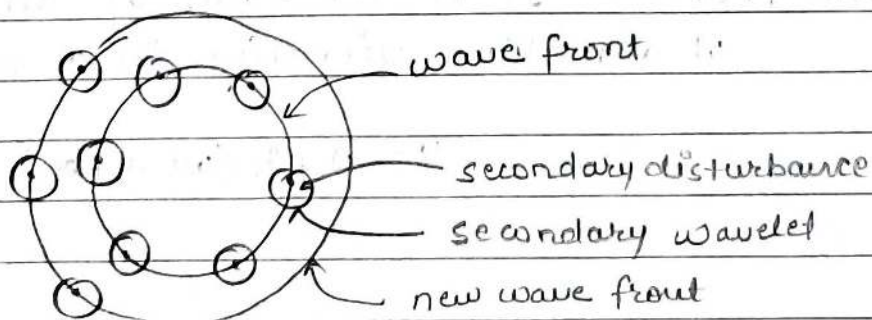
③ Secondary wavelet :-

"Every point on wave front acts as a secondary source of disturbance, emits secondary wavelets."

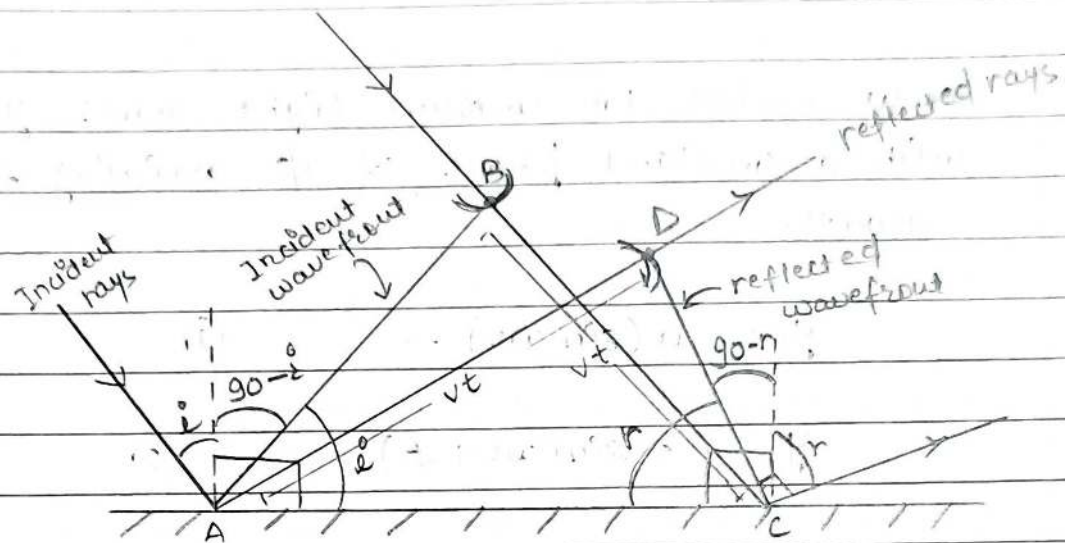


④ Secondary wave front (New wave front) :-

"Common tangent to secondary wavelets drawn in forward direction gives new wave front."

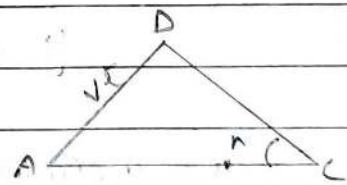
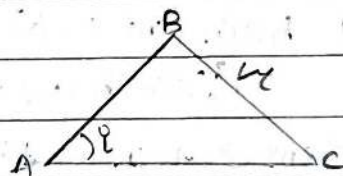


* Proof the law of reflection : ($\angle i = \angle r$)



→ Angle between incident wave front and reflecting surface is angle of incidence (i).

→ Angle between reflected wave front and reflecting surface is angle of reflection (r).



here,

$$BC = AD \text{ (equal length 'vt')}$$

$$\angle B = \angle D \text{ (90°)}$$

$$AC = AC \text{ (common line)}$$

→ According to SAS congruency, both triangles are congruent.

$$\therefore \angle i = \angle r \text{ Proved.}$$

* Superposition of waves and conditions for constructive and destructive interference.

Let us consider two coherent light waves y_1 and y_2 with a constant phase of ϕ arriving from two sources.

$$y_1 = a \sin \omega t \text{ ----- (1)}$$

$$y_2 = b \sin(\omega t + \phi) \text{ ----- (2)}$$

Here, a and b are the amplitudes of waves y_1 and y_2 , respectively.

From the principle of superposition, the resultant wave is given by:

$$y = y_1 + y_2$$

$$y = a \sin \omega t + b \sin(\omega t + \phi)$$

$$[\because \sin(A+B) = \sin A \cos B + \cos A \sin B]$$

$$y = a \sin \omega t + b (\sin \omega t \cos \phi + \cos \omega t \sin \phi)$$

$$y = a \sin \omega t + b \sin \omega t \cos \phi + b \cos \omega t \sin \phi$$

$$y = \sin \omega t (a + b \cos \phi) + b \cos \omega t \sin \phi \text{ ----- (3)}$$

$$\therefore \text{Let } a + b \cos \phi = R \cos \theta \text{ ----- (4)}$$

$$\text{and, } b \sin \phi = R \sin \theta \text{ ----- (5)}$$

Put the (4) and (5) in eqn (3)

$$\therefore y = \sin \omega t R \cos \theta + \cos \omega t R \sin \theta$$

$$y = R [\sin \omega t \cos \theta + \cos \omega t \sin \theta] \quad [\because \sin(A+B)]$$

$$\boxed{y = R \sin(\omega t + \theta)} \text{ ----- (6)}$$

$\therefore$ where, R is resultant amplitude.

* Amplitude of Resultant wave :

from equation (4) and (5)

$$a + b \cos \phi = R \cos \theta \quad \text{--- (4)}$$

$$b \sin \phi = R \sin \theta \quad \text{--- (5)}$$

Squaring and adding equation (4) and (5)

$$(a + b \cos \phi)^2 + (b \sin \phi)^2 = (R \cos \theta)^2 + (R \sin \theta)^2$$

$$\Rightarrow a^2 + b^2 \cos^2 \phi + 2ab \cos \phi + b^2 \sin^2 \phi = R^2 \cos^2 \theta + R^2 \sin^2 \theta$$

$$\Rightarrow a^2 + b^2 \cos^2 \phi + b^2 \sin^2 \phi + 2ab \cos \phi = R^2 \cos^2 \theta + R^2 \sin^2 \theta$$

$$\Rightarrow a^2 + b^2 (\cos^2 \phi + \sin^2 \phi) + 2ab \cos \phi = R^2 (\cos^2 \theta + \sin^2 \theta)$$

$$\Rightarrow R^2 = a^2 + b^2 + 2ab \cos \phi$$

$$\Rightarrow \boxed{R = \sqrt{a^2 + b^2 + 2ab \cos \phi}} \quad \text{--- (7)}$$

for R to be maximum, $\cos \phi$ in (7) should be maximum.

$$\text{i.e., } \cos \phi = +1$$

$$\cos \phi = \cos 2n\pi = 1, \quad [\because n = 0, \pm 1, \pm 2, \dots]$$

$$\Rightarrow \boxed{\phi = 2n\pi = \text{phase difference}} \quad \text{--- (8)}$$

Put this value in equation (7), $\cos \phi = 1$.

$$R = \sqrt{a^2 + b^2 + 2ab \cos \phi}$$

$$R_{\max} = \sqrt{a^2 + b^2 + 2ab} = \sqrt{(a+b)^2}$$

$$\boxed{R_{\max} = a+b} \quad \text{--- (9)}$$

for R to be minimum, $\cos \phi = -1$

$$\cos \phi = \cos (2n+1)\pi = -1 \quad [\because n = 0, \pm 1, \pm 2, \dots]$$

$$\Rightarrow \boxed{\phi = (2n+1)\pi} \text{ ----- (10)}$$

Put this value in eqⁿ (7)

$$R = \sqrt{a^2 + b^2 + 2ab \cos \phi}$$

$$R_{\min} = \sqrt{a^2 + b^2 - 2ab} = \sqrt{(a-b)^2}$$

$$\boxed{R_{\min} = a-b} \text{ ----- (11)}$$

Path difference:

$$\boxed{\text{Path difference } (x) = \frac{1}{2\pi} \phi} \text{ ----- (12)}$$

for maxima

$$\phi = 2n\pi$$

$$\text{Path diff. } (x) = \frac{1}{2\pi} \times 2n\pi$$

$$\boxed{\text{Path difference} = n\lambda}$$

for minima

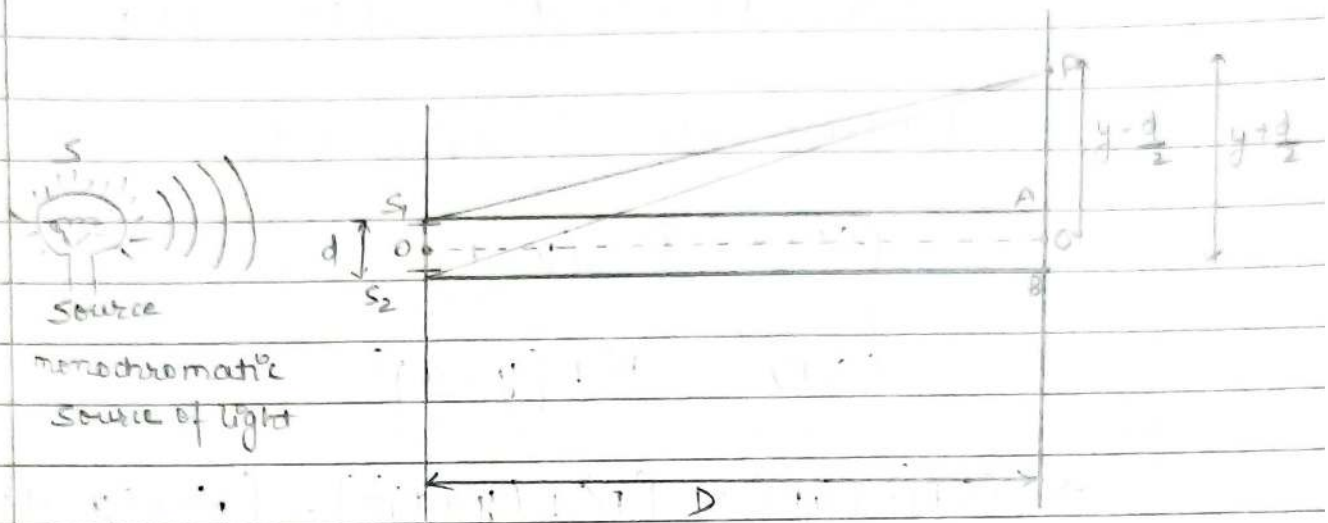
$$\phi = (2n+1)\pi$$

$$\text{Path difference} = \frac{1}{2\pi} (2n+1)\pi$$

$$= \frac{1}{2} (2n+1)\lambda$$

$$\boxed{\text{Path diff.} = \left(\frac{n+1}{2}\right)\lambda}$$

Young's Double slit Experiment



let S_1 and S_2 be two slits separated by a distance 'd' and center 'O' is equidistance from S_1 and S_2

let the slits be illuminated by a monochromatic source of light 'S' of wavelength λ .

The light travelling through slits S_1 and S_2 will meet at point 'P' on the screen which is placed at distance 'O' from slits.

let the path difference between the waves meeting at point P is $= S_2P - S_1P$

i.e., Path difference $= S_2P - S_1P$ ---- (1)

now,

in ΔPS_1A

$$(S_1P)^2 = (S_1A)^2 + (PA)^2 \text{ ---- (2)}$$

In ΔPS_2B

$$(S_2P)^2 = (S_2A)^2 + (PB)^2 \text{ ---- (3)}$$

Since, $S_1A = S_2B = D$

And, $PA = y - \frac{d}{2}$ and $PB = y + \frac{d}{2}$

Put those values in eqⁿ (2) and (3)

$$(S_1P)^2 = D^2 + \left(y - \frac{d}{2}\right)^2$$

$$(S_2P)^2 = D^2 + \left(y + \frac{d}{2}\right)^2$$

$$\Rightarrow (S_2P)^2 - (S_1P)^2 = \left[D^2 + \left(y + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(y - \frac{d}{2}\right)^2\right]$$

$$\Rightarrow (S_2P)^2 - (S_1P)^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$\Rightarrow (S_2P + S_1P)(S_2P - S_1P) = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

$$\Rightarrow \frac{S_2P - S_1P}{S_2P + S_1P} = \frac{y^2 + \frac{d^2}{4} + yd - y^2 - \frac{d^2}{4} + yd}{S_2P + S_1P}$$

$$\Rightarrow \frac{S_2P - S_1P}{S_2P + S_1P} = \frac{2yd}{S_2P + S_1P} \quad \text{--- (4)}$$

If point P is very close to O, then

$$S_2P \approx D \text{ and } S_1P \approx D$$

$$\Rightarrow S_2P + S_1P \approx 2D$$

Put this value in eq (4)

$$S_2P - S_1P = \frac{2yd}{2D} = \frac{yd}{D}$$

$$\boxed{\text{Path difference} = S_2P - S_1P = \frac{yd}{D}} \quad \text{--- (5)}$$

✱ for Constructive interference (Bright fringes)

$$\text{Path difference} = n\lambda \quad [\because n = 0, \pm 1, \pm 2, \dots]$$

$n = \text{integer.}$

using (5) $\frac{y \cdot d}{D} = n\lambda$

$$\Rightarrow \boxed{y_{nd} = \frac{n\lambda D}{d}}$$

Here, y_n is the distance of n^{th} bright fringe from centre.

✱ for Destructive interference (Dark fringe)

$$\text{Path difference} = (2n+1) \frac{\lambda}{2}$$

using (5) $\frac{y d}{D} = (2n+1) \frac{\lambda}{2}$

$$\boxed{y_n = \frac{(2n+1) \lambda D}{2d}}$$

or

$$\boxed{y_{nd} = \left(n + \frac{1}{2}\right) \frac{\lambda D}{d}} \quad n = 0, \pm 1, \pm 2, \dots$$

Here y_{nd} is the distance of n^{th} dark fringe from center.

✱ fringe width (β) :

→ Distance between two adjacent bright (or dark) fringes is called the fringe width.

for bright fringes

$$\text{fringe width } (\beta) = y_{n+1} - y_n$$

$$= (n+1) \frac{\lambda D}{d} - n \frac{\lambda D}{d}$$

$$\boxed{\beta = \frac{\lambda D}{d}} \quad \text{--- (A)}$$

Dark fringes

$$\text{fringe width } (\beta) = y_{(n+1/2)} - y_{n/2}$$

$$= \left(n+1+\frac{1}{2}\right) \frac{\lambda D}{d} - n \frac{\lambda D}{d} \left(n+\frac{1}{2}\right) \frac{\lambda D}{d}$$

$$= \left(n+\frac{3}{2}\right) \frac{\lambda D}{d} - \left(n+\frac{1}{2}\right) \frac{\lambda D}{d}$$

$$\boxed{\beta_d = \frac{\lambda D}{d}} \quad \text{--- (B)}$$

Here the fringe width (β) for bright and dark fringes is equal.

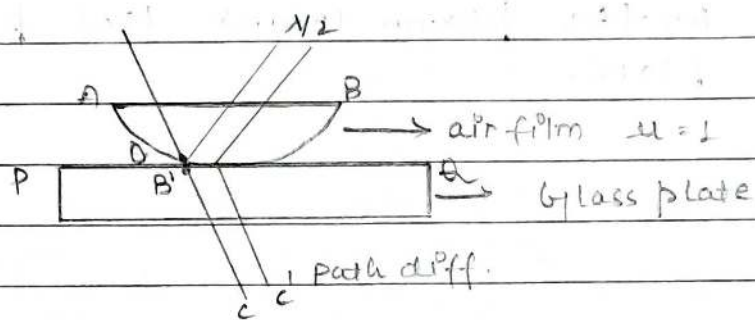
i.e.,

$$\boxed{\beta = \frac{\lambda D}{d}}$$



Newton's Rings

- Newton's Rings is a phenomenon in which an interference pattern is created by the reflection of light between two surfaces - a plano convex lens and an adjacent touching glass plate. It is named after ISAAC Newton who investigated the effect in 1704.



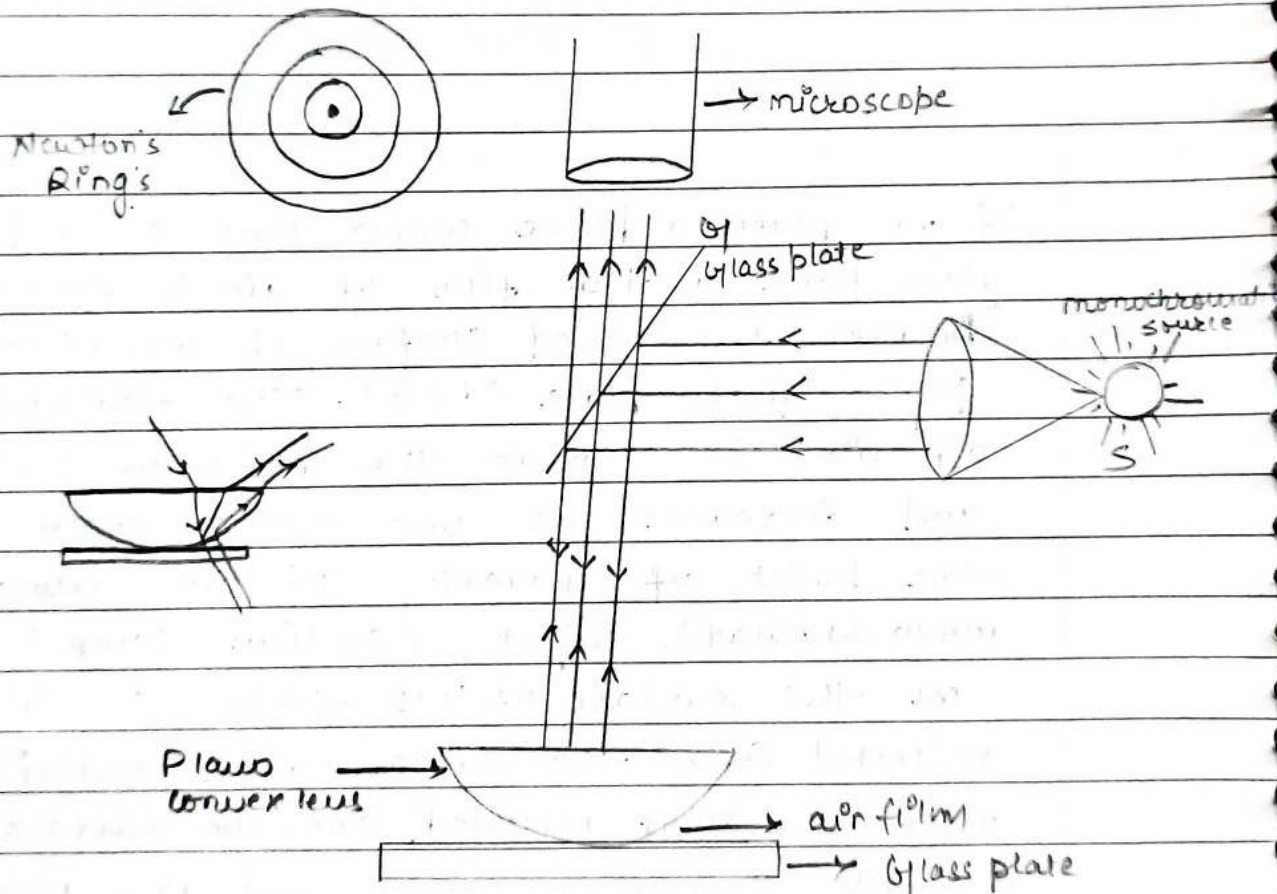
If we place a plano convex lens on a plane glass plate a thin film of air is formed between the curved surface of lens (AOB) and plane glass plate (POQ). The thickness of air film is zero at the point of contact 'O' and increases as one moves away from the point of contact. If we allow monochromatic light (sodium lamp) to fall on the surface of lens then the light reflected from the surface AOB interference with the light reflected from the surface POQ, to give concentric bright and dark rings called Newton's Rings.

Ques. why ring is formed?

Ans:- The locus of constant thickness of film lies on a circle.

Experimental Setup:

- It consists of a monochromatic light which is incident on a convex lens to make parallel beam of light. This parallel beam is further incident on a glass plate G , inclined at 45° to the beam. The plate reflects the beam downwards which in turn is incident normally on the plano convex lens placed on a glass plate.



* Newton Rings by reflected light

(E) when the reflection of light is occurring while going from denser to a rarer medium (eg. air), a path difference of $\frac{1}{2}$ corresponding to a phase difference of ' π ' is introduced.

(ii) further, in case of thin film interference

$$\text{path difference } (\Delta) = 2ut \cos r$$

u = refractive index of thin film

$u = 1$ for air thin film

t = thickness of air film

r = angle of refraction

for normal incidence $r = 0^\circ$

$$\cos r = \cos 0^\circ = 1$$

Hence, path difference for thin film of air $= 2 \cdot 1 \cdot t \cos 0 = 2t$

$$\text{Total path difference } \Delta = \frac{1}{2} + 2t \quad \text{--- (1)}$$

* for Constructive Interference (Bright Rings)

$$\text{Path difference } (\Delta) = n\lambda$$

$$2t = n\lambda - \frac{1}{2} = \left(n - \frac{1}{2}\right) \lambda$$

$$2t = (2n - 1) \frac{\lambda}{2} \quad \text{--- (2)}$$

* for destructive interference (Dark fringes).

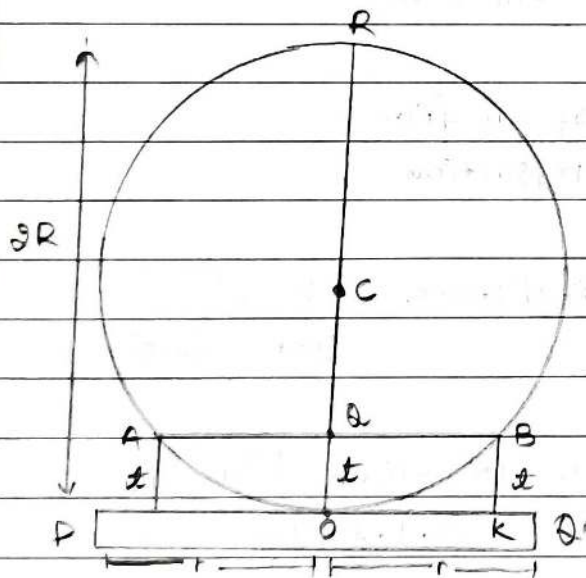
$$\text{Path difference } (\delta) = \left(n + \frac{1}{2}\right) \lambda \quad [\because n = 0, \pm 1, \pm 2, \dots]$$

using eq (1) $\frac{\lambda}{2} + 2t = \left(n + \frac{1}{2}\right) \lambda$

$$\frac{\lambda}{2} + 2t = n\lambda + \frac{\lambda}{2}$$

$$\boxed{2t = n\lambda} \quad \text{--- (3)}$$

* Diameter of Newton Ring's



let the radius of curvature of lens be 'R'
 let us take a point K on the plate such that the thickness of the film at it be 't'. All such points will lie on a circle with O as the centre and r as the radius.

By Geometry of circle

$$AO \times OB = OQ \times OR$$

$$AO \times OB = OQ \times (OR - OQ) \quad \text{--- (4)}$$

Here, $AB = OB = r$, $OA = t$, $OR = 2R$

Put these value in eqⁿ (4)

$$r^2 = t(2R - t)$$

$$\Rightarrow r^2 = 2Rt - t^2$$

If thickness of air is very small, then $t^2 \ll L$

So, t^2 can be neglected.

$$\Rightarrow r^2 = 2Rt$$

$$\Rightarrow r = \sqrt{2Rt}$$

$$\Rightarrow r = \sqrt{(2t)R} \quad = \text{radius of ring.}$$

$$\text{Diameter of ring } d = 2r = \sqrt{(2t)R}$$

$$d = \sqrt{4(2t)R}$$

* Diameter of dark Ring

using eqⁿ (3),

$$d_n = \sqrt{4n\lambda R} \rightarrow \text{diameter of } n^{\text{th}} \text{ dark Ring.}$$

for $n = 0$

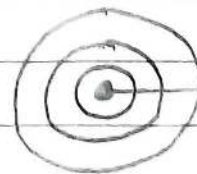
$$d_0 = 0 \Rightarrow \text{we get dark ring at the centre in Newton Rings formed by Reflection.}$$

* Diameter of Bright Ring

using eqⁿ (2)

$$d_n = \sqrt{4(2n+1)\frac{\lambda}{2}}$$

$$\Rightarrow d_n = \sqrt{2(2n+1)\lambda} \rightarrow \text{diameter of } n^{\text{th}} \text{ bright Ring.}$$



dark ring at the centre.

$$\Rightarrow r^2 = 2Rt = t^2$$

* thickness of air is very

* Newton's Rings by Transmitted light

→ for transmitted lights, there is no path difference between the superimposing transmitted waves.

Only path difference introduced is due to thin film of air.



Path difference (Δ) = $2t$ (for thin film of air)

* for constructive interference (^{Bright} ~~dark~~ ring)

Path difference $\boxed{2t = n\lambda}$

* for destructive interference (dark ring)

Path difference $\boxed{2t = \left(n + \frac{1}{2}\right)\lambda}$

Since,

$$\text{diameter } d = \sqrt{4(2t)R}$$

* Diameter of dark ring.

$$\boxed{d_n = \sqrt{4n\lambda R}}$$

$$\boxed{d_n \propto \sqrt{n}}$$

* Diameter of Bright Ring

$$d_n = \sqrt{4\left(n + \frac{1}{2}\right)R\lambda} = \sqrt{(2n+1)2R\lambda}$$

$$\boxed{d_n \propto \sqrt{2n+1}}$$

→ Dark ring in transmitted system is bright. So, central spot where is dark reflected system will be bright in transmitted system.

* Uses of Newton Rings:

(i) Determination of wavelength.
for dark rings

diameter of n^{th} dark ring $d_n = \sqrt{4n\lambda R}$

$$\Rightarrow d_n^2 = 4n\lambda R$$

m^{th} dark ring $d_m^2 = 4m\lambda R$

$$\Rightarrow d_m^2 - d_n^2 = 4\lambda R (m-n)$$

$$\Rightarrow \boxed{\lambda = \frac{d_m^2 - d_n^2}{4R(m-n)}}$$

R = radius of curvature of Plano-convex lens.

(ii) Determination of refractive index of a liquid.

instead of air film with $\mu=1$, we are having a film of liquid with refractive index μ , then,

diameter of n^{th} dark ring $d_n = \sqrt{\frac{4n\lambda R}{\mu}}$

for m^{th} dark ring

$$d_m^2 = \frac{4m\lambda R}{\mu}$$

$$\Rightarrow d_m^2 - d_n^2 = \frac{4\lambda R}{\mu} (m-n)$$

$$\Rightarrow \boxed{\mu = \frac{4\lambda R (m-n)}{d_m^2 - d_n^2}}$$

* **Interference** :- The phenomenon of redistribution of light energy due to the superposition of light waves from two or more coherent sources is known as interference.

→ There are two methods of producing interference:

(i) **Division of wavefront** :-

→ In this case, a wavefront coming from a narrow slit is divided by passing it through two closely spaced slits. The waves coming through these two slits recombine on screen and give fringes pattern.

(ii) **Division of amplitude** :- the amplitude of light of the

→ In this case, the wave is divided into two parts, namely reflected and transmitted components by partial reflection at the surface. The two parts travel through

different paths and recombines to produce interference pattern.

* Conditions of sustained interference .

- (i) Two sources of light must be coherent i.e, these must have same wavelength or frequency.
- (ii) These must have constant phase difference.
- (iii) The sources of light must have same amplitude.

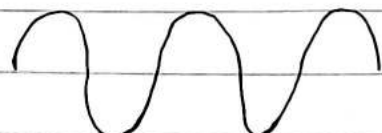
* **wave front :-** It is a set (locus) of all points that are in phase.

* **Huygen Principle :-** All points of a wave front are the new source of secondary wavelets that expand in every direction.

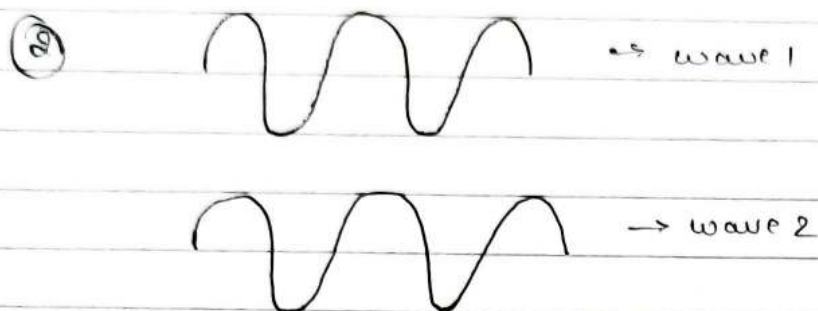
* Coherent Sources of light :

→ If two or more waves maintain same phase or phase difference over a distance and time then, they are said to be coherent.

eg - ①  → wave 1

 → wave 2

wave 1 and 2 are in phase, so, these are coherent waves.

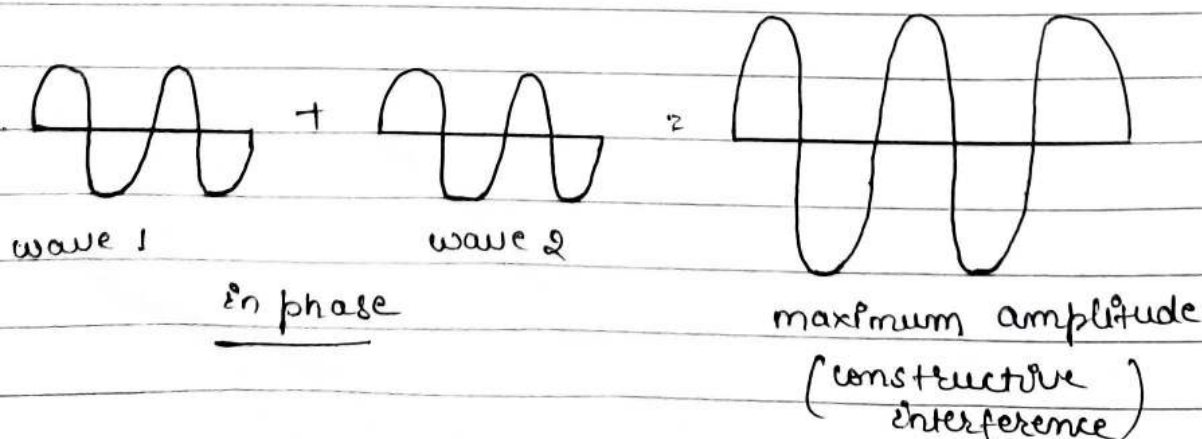


Here, also wave 1 and wave 2 are coherent waves because phase difference of 90° is constant with time.

further, coherent waves / sources of light have also same frequency, same wavelength, almost same amplitude and have no phase difference or constant phase difference over time.

* Constructive Interference:

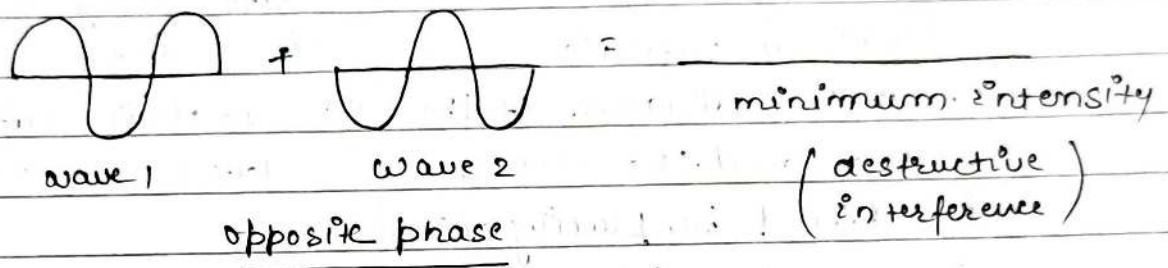
→ If two waves superimpose with each other in the same phase, the amplitude of the resultant wave is equal to the sum of the amplitude of the individual waves, resulting in the maximum intensity is known as Constructive Interference.





Destructive Interference:

→ If two waves superimpose each other in opposite phase, then the superposition of that waves results in the minimum intensity of light, known as destructive interference.



Difference between Diffraction and Interference.

Interference	Diffraction
①. Interference is the result of interaction of light coming from two different wavefronts originated from the same source.	①. Diffraction pattern is the result of interaction of light coming from different parts of the same wavefront.
②. Interference fringes may or may not be of the same width.	②. Diffraction fringes are not of the same width.
③. The intensity of maxima is same.	③. The intensity of maxima is not same.
④. All bright bands are of uniform intensity.	④. All bright bands are not of same intensity.
⑤. Point of minimum intensity are perfectly dark.	⑤. Points of minimum intensity are not perfectly dark.



Difference between Fraunhofer diffraction and Fresnel's diffraction.

Fraunhofer diffraction	Fresnel's diffraction
①. The distance of the source and the screen from diffracting element is effectively infinite.	①. The distance between the source or screen or both from diffracting obstacle is finite.
②. The infinite distance of the source and the screen is achieved by placing them in the focal plane of two convex lens.	②. In this diffraction no lenses are necessary.
③. The incident wavefront is plane.	③. The incident wavefront is divergent either spherical or cylindrical.
④. The theoretical treatment is much simpler.	④. The theoretical treatment is more complex and only approximate.
⑤. The centre of the diffracting pattern is always bright because all paths are parallel to the axis of the lens.	⑤. The centre of diffraction pattern may be bright or dark depending upon the number of Fresnel's zones.

* Fraunhofer diffraction from a single slit:

- ①. Condition of central maxima.
- ②. Condition of minima.
- ③. Condition of principle maxima, secondary - maxima.

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} \rightarrow 1$$

$$\alpha = \frac{\pi a \sin \theta}{\lambda}$$

- ④. Condition of principle of secondary maxima.

→ For maximum, value, $\sin \alpha$ or α must have its maximum value.

For α maximum →

$$a \sin \theta = \pm \left(m + \frac{1}{2} \right) \lambda$$

$$\alpha = \frac{\pi a \sin \theta}{\lambda} = \pm \frac{\pi}{\lambda} \times \left(m + \frac{1}{2} \right) \lambda$$

$$\alpha = \pm \left(m + \frac{1}{2} \right) \pi$$

$$m = +1, \alpha = \frac{3}{2} \pi$$

$$m = 2, \quad \alpha = \frac{5\pi}{2}$$

$$m = 3, \quad \alpha = \frac{7\pi}{2}$$

So,

for $m = 1, 2, 3, \dots$

$$\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}$$

$$\text{when } \alpha = \pm \frac{3\pi}{2}$$

$$I_{\alpha = \pm \frac{3\pi}{2}} = I_0 \left[\frac{\sin\left(\frac{3\pi}{2}\right)^2}{\frac{3\pi}{2}} \right] \approx \frac{4}{9\pi^2} I_0$$

$$I_{\alpha = \pm \frac{5\pi}{2}} = I_0 \left(\frac{\sin 5\pi/2}{5\pi/2} \right) \approx \frac{4}{25\pi^2} I_0$$

$$I_{\alpha = \pm \frac{7\pi}{2}} = I_0 \left(\frac{\sin 7\pi/2}{7\pi/2} \right) \approx \frac{4}{49\pi^2} I_0$$

Q. * Prove that in Fraunhofer diffraction from a single slit, the Ratio of intensity is given by.

$$1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} \approx 1 : \frac{1}{22} : \frac{1}{61} : \frac{1}{121}$$

To determine the position of principle or secondary maxima.

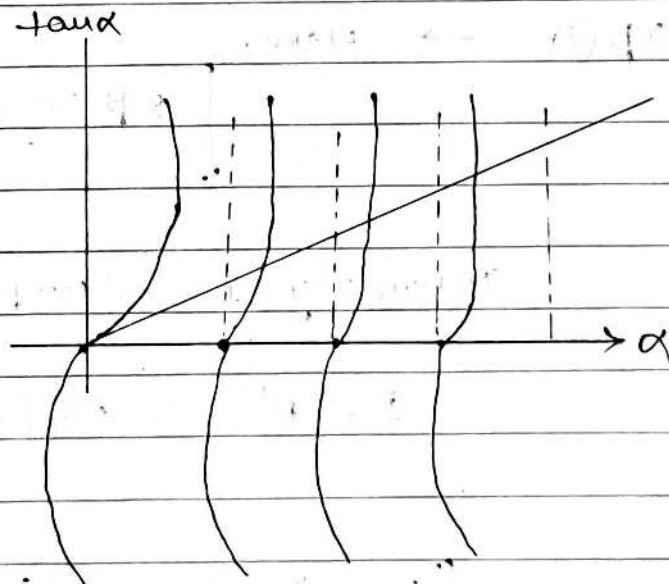
$$\frac{dI}{d\alpha} = 0$$

$$\frac{d}{d\alpha} I_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 = 0$$

$$I_0^2 \frac{d}{d\alpha} \left(\frac{\sin \alpha}{\alpha} \right)^2 = 0$$

$$I_0^2 2 \left(\frac{\sin \alpha}{\alpha} \right) \frac{d}{d\alpha} \left(\frac{\sin \alpha}{\alpha} \right) = 0$$

$$I_0^2 2 \left(\frac{\sin \alpha}{\alpha} \right) \cdot \frac{(\alpha \cos \alpha - \sin \alpha)}{\alpha^2} = 0$$



* Plane - diffraction grating. (N-slits)

Grating :- N-parallel slits placed together from a grating.

Ques: Define grating elements?

Ans:- The sum of width of transparent (a) and opaque region (b) is called grating element.

$$\text{grating element} = a + b$$

$$\text{Single slit} \Rightarrow R = A_0 \frac{\sin \alpha}{\alpha}$$

Resultant amplitude 'R' is given by

$$\text{Step (1)} \rightarrow R' = A_0 \frac{\sin \alpha}{\alpha} \cdot \frac{\sin N\beta}{\beta}$$

β = phase difference between light waves diffracted from N-slits.

Step (2) $\rightarrow$ Here,

$$2\beta = \frac{2\pi}{\lambda} (a+b) \sin \theta$$

$$\text{Intensity } I = (\text{Amplitude})^2$$

$$I = R'^2 = A_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \left(\frac{\sin N\beta}{\beta} \right)^2$$

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N\beta}{\beta} \right)^2$$

$$I_0 = A_0^2$$



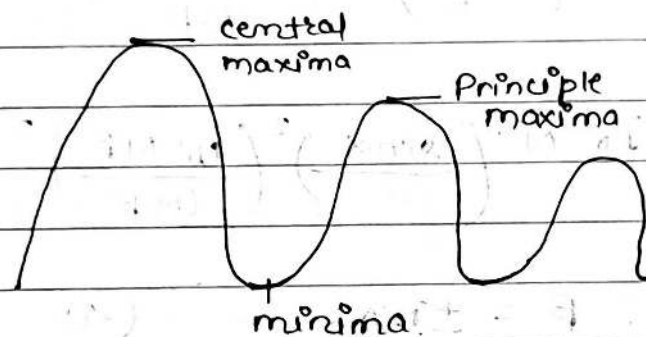
Diffraction Grating (n slits)

$$\text{Path difference} = (a+b) \sin \theta \quad \text{--- (1)} \quad \left[\begin{array}{l} \text{single slit} = a \sin \theta \\ \text{path diff.} \end{array} \right]$$

$a+b = \text{grating element}$

$$\text{phase difference} = 2\pi \quad \text{--- (2)}$$

$$\text{Intensity (I)} = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \left(\frac{\sin N\beta}{\sin \beta} \right)^2 \quad \text{--- (3)}$$



For Central Maxima.

for minimum Intensity

$$\sin \beta = 0 \quad \text{---} \quad \sin n\pi \quad [n = 0, \pm 1, \pm 2, \dots]$$

$$\Rightarrow \boxed{\beta = n\pi} \rightarrow \text{maxima}$$

$$n = 0 \rightarrow \text{central maxima}$$

$$n = 1 \rightarrow \text{1st order principle maxima}$$

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

$$n = 0, \quad \beta = 0$$

$$I = \frac{0}{0} \text{ undefined}$$

$$\text{If } \frac{f(x)}{g(x)} = \frac{0}{0} \text{ at } x = x_0$$

then by L. Hospital Rule

$$\frac{f(x)}{g(x)} = \frac{f'(x)}{g'(x)}$$

$$\lim_{\beta \rightarrow \pm n\pi} I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N\beta}{\sin \beta} \right)^2 = \frac{0}{0} \text{ undefined.}$$

using L'Hospital Rule.

$$\frac{\sin N\beta}{\sin \beta} = \frac{N \cos N\beta}{\cos \beta}$$

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{N \cos N\beta}{\cos \beta} \right)^2$$

$$I = I_0 N^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\cos N\beta}{\cos \beta} \right)^2$$

$$\therefore \beta = \pm n\pi \quad (4)$$

$n = 0$, I_0 is maximum value

$$I = I_0 N^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 = \text{Maximum value at the centre.}$$

for $n = \pm 1, \pm 2, \dots$ we will get principle maxima.

* for Minima.

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

for minima $I = 0$

$$\text{i.e., } \sin N\beta = 0 = \sin m\pi \quad [\because m = 0, \pm 1, \pm 2, \dots]$$

$$\Rightarrow N\beta = m\pi$$

$$\boxed{\beta_{\min} = \frac{m\pi}{N}} \quad \text{--- (5)} \quad [\because N = \text{Number of slits}]$$

*

$$\text{Path difference} = \frac{\lambda}{2\pi} \times \text{phase difference.}$$

$$(a+b) \sin \theta = \frac{\lambda}{2\pi} \times 2\beta$$

$$(a+b) \sin \theta = \frac{\lambda \beta}{\pi} \quad \text{--- (6)}$$

for maxima, $\beta = \pm n\pi$

Put this value in eqⁿ (6)

$$(a+b) \sin \theta = \frac{\lambda}{\pi} \times (\pm n\pi)$$

$$\boxed{(a+b) \sin \theta = n\lambda} \quad \text{--- (7)}$$

This is the condition to get principle maxima.

✱

for minima:

Path difference.

$$(a+b) \sin \theta = \frac{\lambda}{2\pi} (2\beta)_{\min}$$

$$(a+b) \sin \theta = \frac{\lambda}{\pi} \beta_{\min} = \frac{\lambda}{\pi} \left(\pm \frac{m\pi}{N} \right)$$

$$N(a+b) \sin \theta = \pm m \lambda \quad \text{--- (8)}$$

Put the value of $(a+b) \sin \theta$ from (7) in L.H.S of (8)

$$Nn \lambda = m \lambda$$

$$m = nN$$

$$n = 0, \pm 1, \pm 2, \dots \rightarrow \text{maxima.}$$

$$m = 0, \pm N, \pm 2N, \dots \rightarrow \text{we will not get minima.}$$

Revsion

↓

$$(a+b) \sin \theta = \text{path diff.}$$

$$2\beta = \text{phase diff.}$$

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N\beta}{\sin \beta} \right)$$

Maxima, $\sin \beta = 0$

Minima $I = 0$ $\sin N\beta = 0$

$$\sin m\pi$$

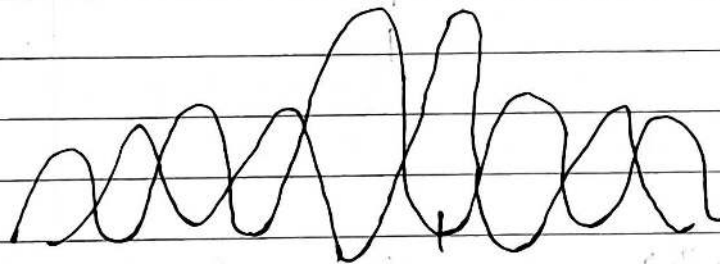
$$m = 0, \pm 1, \dots$$

* Resolving Power

- The ability of an optical instrument to form images of separate image of two closely spaced objects is called Resolving power.

* Rayleigh Criteria.

- According to Rayleigh criteria two closely spaced objects are seen just separate when first minima of first object falls on the central first maxima of second object.



* Resolving power of diffraction grating:

- It is the ability of grating to form separate diffraction maxima of two wavelengths which are very close to each other.

- It is denoted as $\frac{1}{d\lambda}$

where, $d\lambda$ is the difference between two wavelengths

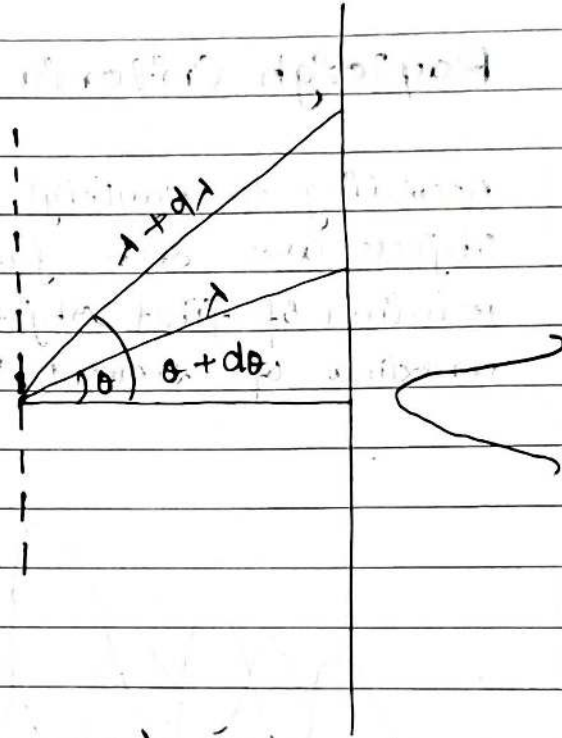
eg - $\lambda_1 = 5298 \text{ \AA}$

$\lambda_2 = 5298.6 \text{ \AA}$

$\therefore d\lambda = .6 \text{ \AA}$

Derivation to find out resolving power of diffraction grating.

let the wavelength λ is diffracted by using angle θ and $(\lambda + d\lambda)$ is diffracted by angle $(\theta + d\theta)$



The condition of maximum for wavelength ' λ '

$$(a+b) \sin \theta = n \lambda \quad \text{--- (1)}$$

condition of minima is

$$N(a+b) \sin \theta = m \lambda \quad \text{--- (2)}$$

$n = \text{for maxima}$
 $m = \text{minima}$
 $N = \text{No. of slits}$

There are $m = nN + 1$ minima for 'n' principle maxima.

To remember:

→ The first minimum for in direction $\theta + d\theta$ is

$$N(a+b) \sin \theta = (nN + 1)$$

$$N(a+b) \sin(\theta + d\theta) = (nN + 1) \lambda \quad \text{--- (3)}$$

for wavelength $(\lambda + d\lambda)$, the condition of maxima is

$$(a+b) \sin(\theta + d\theta) = n(\lambda + d\lambda) \text{ --- (4)}$$

Substitute value of L.H.S of (4) in L.H.S of (3)

$$Nn(\lambda + d\lambda) = nN\lambda + 1$$

$$Nn\lambda + Nnd\lambda = nN\lambda + 1$$

$$Nnd\lambda = 1$$

$$\boxed{\frac{1}{d\lambda} = nN}$$

Resolving power
of grating.

To Remember:

- ① $\lambda \rightarrow$ first minima
replace θ by $d + d\theta$
- ② $\lambda + d\lambda \rightarrow$ Principal maxima
 $\theta + d\theta$
- ③ first minima of $\lambda =$ Principal maxima of
According to Rayleigh criteria.

Ques:-

Ans:-

What are the type of plane polarized light?

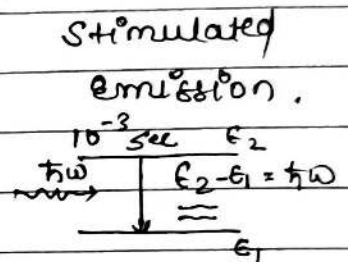
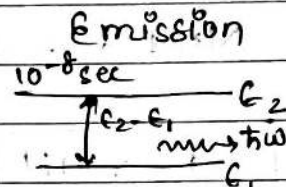
There are three type of polarized light.

- ① Plane polarized light
- ② Circularly polarized light
- ③ Elliptically polarized light

Date : _____

①.	Absorption	Emission
----	------------	----------

Spontaneous



Ques. Define meta stable states.

→ The state having higher lifetime $\sim 10^{-3}$ sec.

(6 Marks)
Answer

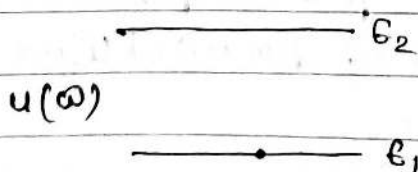
Einstein coefficients:

$E = h\nu$

h cut

A hand-drawn diagram of a cell, represented by an oval. Inside the oval, there are two labels: 'h' and '2h'. The label 'h' is positioned above the label '2h'. There are also some faint, illegible markings around the oval.

①. Stimulate Absorption

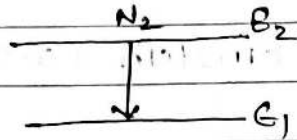


$n(\omega)$ = Density of photon

$$N_1 = \text{No. of atoms in energy state } E_1$$

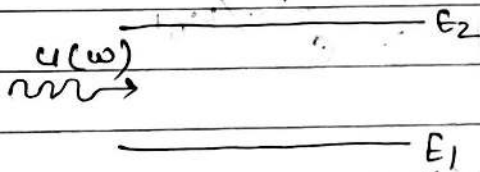
Rate of stimulated absorption $\propto N_1 u(\omega)$
 $= B_{12} N_1 u(\omega) \text{ --- (1)}$

(2). Spontaneous Emission.



Rate of spontaneous emission $\propto N_2$
 $= A_{21} N_2 \text{ --- (2)}$

(3). Stimulated Emission.



Rate of stimulate Emission $\propto N_2 u(\omega)$
 $= B_{21} N_2 u(\omega) \text{ --- (3)}$

At equilibrium,

Rate of emission = Rate of Absorption.

$$A_{21} N_2 + B_{21} N_2 u(\omega) = B_{12} N_1 u(\omega)$$

$$\Rightarrow A_{21} N_2 = (B_{12} N_1 - B_{21} N_2) u(\omega)$$

$$\Rightarrow u(\omega) = \frac{A_{21} N_2}{B_{12} N_1 - B_{21} N_2}$$

$$\Rightarrow u(\omega) = \frac{A_{21} N_2}{B_{21} N_2 \left(\frac{B_{12}}{B_{21}} \frac{N_1}{N_2} - 1 \right)}$$

$$u(\omega) = \frac{A_{21}}{B_{21} \left(\frac{B_{12}}{B_{21}} \frac{N_1}{N_2} - 1 \right)}$$

A_{21}, B_{12}, B_{21} = Einstein Co-efficients.

No. of atoms (N_i) energy state E_i are given by:

$$\frac{N_2}{N_1} = \frac{e^{-E_2/k_B T}}{e^{-E_1/k_B T}}$$

$$\frac{N_2}{N_1} = \frac{e^{-E_2/k_B T}}{e^{-E_1/k_B T}}$$

$$N_1 = \frac{e^{-E_1/k_B T}}{e^{-E_1/k_B T}}$$

$\therefore k_B$ = Boltzmann constant.

$$N_2 = \exp\left(\frac{-E_2}{k_B T}\right) = e^{-E_2/k_B T}$$

$$\frac{N_1}{N_2} = \frac{e^{-E_1/k_B T}}{e^{-E_2/k_B T}}$$

$$\frac{N_1}{N_2} = e^{-E_1/k_B T} \cdot e^{E_2/k_B T}$$

$$\frac{N_1}{N_2} = e^{\left(\frac{-E_1}{k_B T} + \frac{E_2}{k_B T} \right)}$$

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/k_B T}$$

Put this value in eq (1).

$$u(\omega) = \frac{A_{21}}{\frac{B_{12}}{B_{21}} e^{(E_2 - E_1)/k_B T} - 1}$$

$$u(\omega) = \frac{A_{21}}{B_{21}} \left(\frac{B_{12}}{B_{21}} e^{\hbar\omega/k_B T} - 1 \right) \quad \text{--- (2)}$$

According to Planck's law of radiation,

$$u(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \left(\frac{e^{\hbar\omega/k_B T}}{e^{\hbar\omega/k_B T} - 1} \right) \quad \text{--- (3)}$$

Comparing (2) and (3).

$$\frac{B_{12}}{B_{21}} = 1 \Rightarrow \boxed{B_{12} = B_{21}}$$

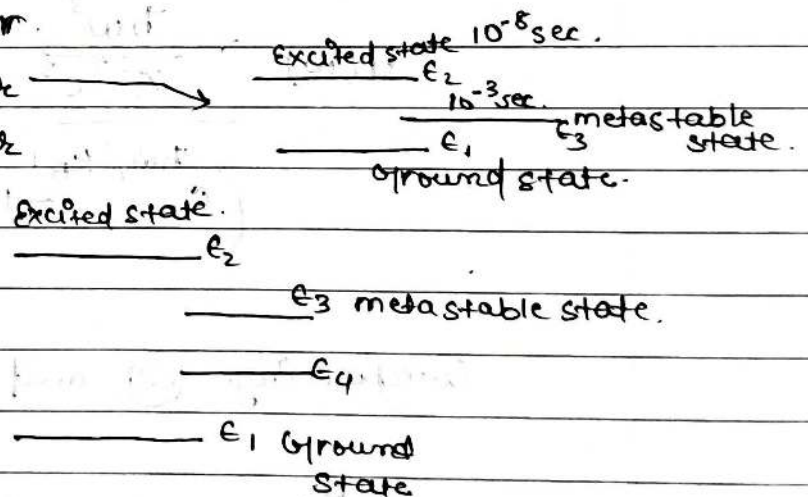
$$\Rightarrow \frac{A_{21}}{B_{21}} = \frac{\hbar\omega^3}{\pi^2 c^3}$$

- ① Types of laser
- ② Population Inversion
- ③ Components of laser
- ④ Types of pumping
- ⑤ He-Ne laser
- ⑥ Properties of laser
- ⑦ Ruby laser.
- ⑧ Meta stable
- ⑨ Einstein co-efficient.
- ⑩ Spontaneous emission.
- ⑪ Stimulated emission.
- ⑫ Stimulated absorption.

① Types of laser.

① Three-level laser

② Four-level laser



* Properties of laser.

- ① High intensity
- ② uni-directional
- ③ spread of light is very very less.

Population inversion

→ The process of accumulation of a very large number of atoms in excited state as compared to ground state is called population inversion.

Ques. what is the basic phenomenon behind the laser?

Components of laser

(1) Active medium.

(2) Pumping

(3) { Cavity
→ Resonant

Types of Pumping

(1) optical pumping (light photons) are used to pump atoms from low to high energy state. (Ruby laser)

(2). Electrical pumping - electrons are used for pumping. (He-Ne laser)

(3). Direct conversion.

V.V.V.S

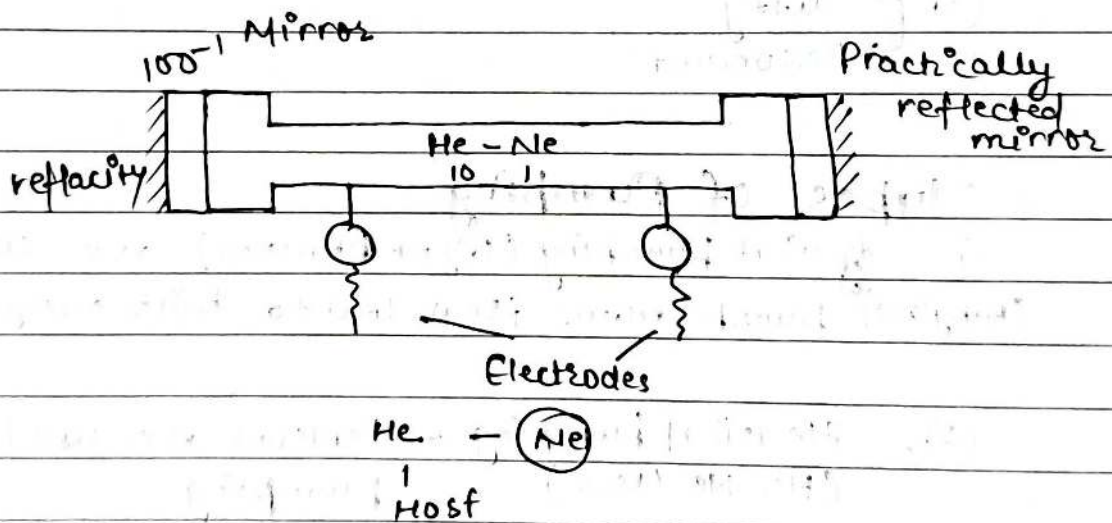
6 marks

He - Ne laser

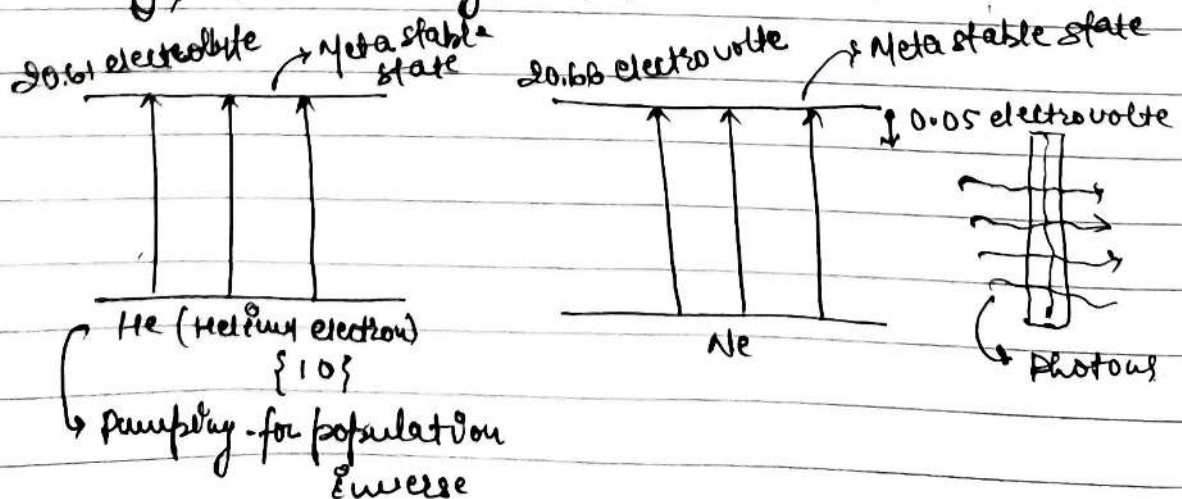
Characteristics

- ① Gas laser
- ② He : Ne = 10 : 1 $\rightarrow$ active medium
- ③ 4-level laser
- ④ Electric pumping is used.
- ⑤ Very High Efficiency
- ⑥ Continuous laser light of red colour with $\lambda = 6328 \text{ \AA}$ is produced.

Diagram



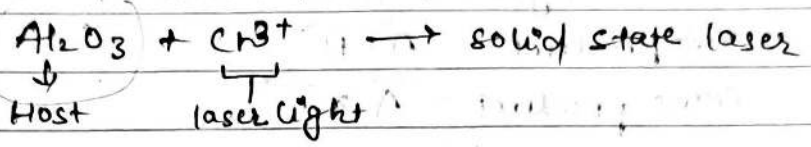
Energy level diagram :



0.05
(5 marks)

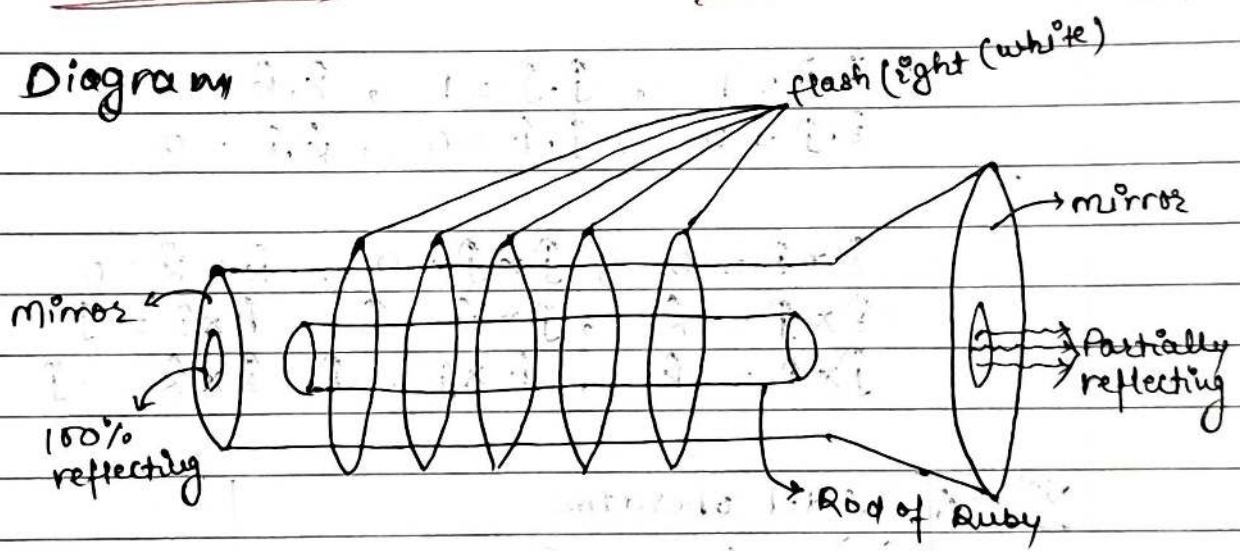
Ruby laser

Active Ruby (Al_2O_3)



- Ruby crystal consist of 0.05% chromium by weight.
- Optical pumping $\rightarrow$ in use white flash light. (green color)

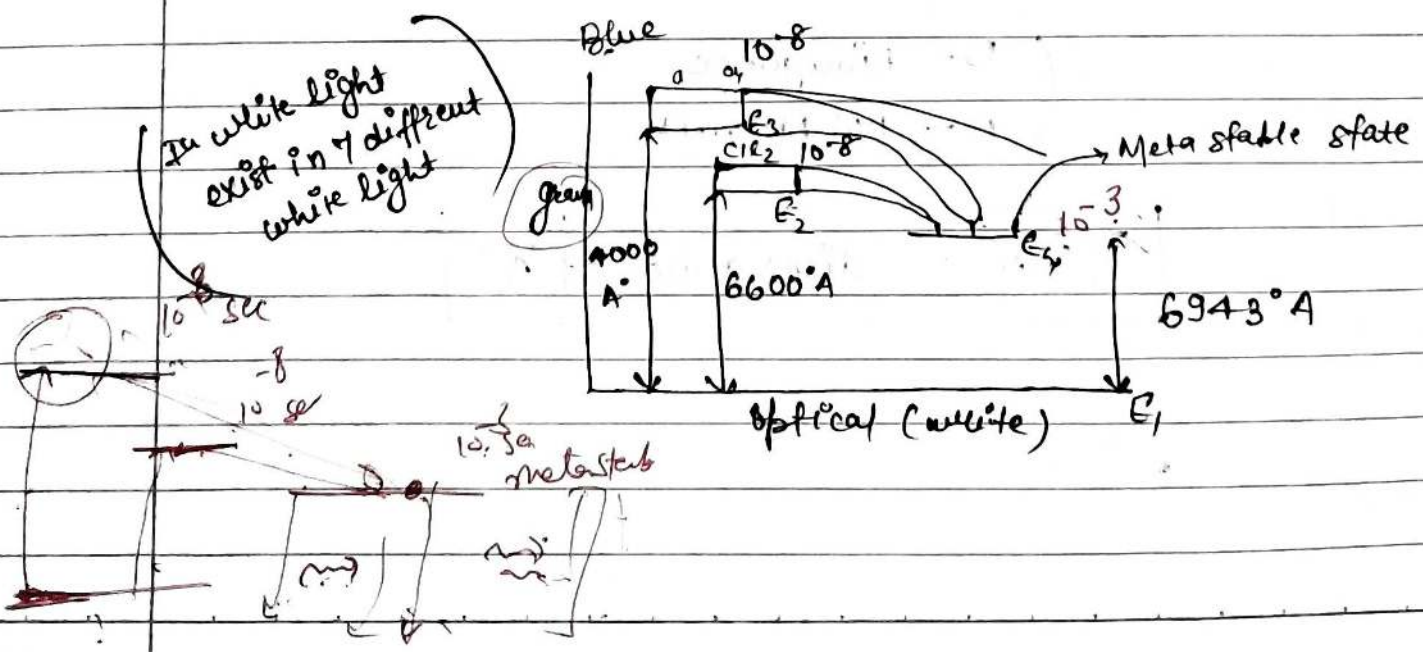
Diagram



Energy level diagram

$E = h\nu$

4-level laser



Unit - II part - A

Date : _____

$\vec{A}$ = vector

$\vec{E}$ = Electric field vector

$\vec{B}$ = Magnetic field vector

dot product = $\vec{A} \cdot \vec{B}$

Cross product = $\vec{A} \times \vec{B}$

$$\vec{A} = A\hat{i} + B\hat{j} + C\hat{k}$$

$\hat{i}, \hat{j}, \hat{k}$ = unit vectors

$$\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0$$

$$\hat{i} \times \hat{i} = 0, \hat{j} \times \hat{j} = 0, \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$$

$$\hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}$$

U.V.I

∇ = del operator

$$\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Divergence of any vector field $\vec{A}$

(D - Divergence,
D - Dot product)

$$\vec{\nabla} \cdot \vec{A} = \text{divergence of } \vec{A}$$

curl of any vector field $\vec{A}$

(C - curl
C - cross product)

$$\vec{\nabla} \times \vec{A} = \text{curl of } \vec{A}$$

$\vec{E}$ = Electric field

divergence of electric field $\vec{E} = \vec{\nabla} \cdot \vec{E}$

curl of electric field $\vec{E} = \vec{\nabla} \times \vec{E}$

divergence of magnetic field $\vec{B} = \vec{\nabla} \cdot \vec{B}$

curl of magnetic field $\vec{B} = \vec{\nabla} \times \vec{B}$

$\therefore \int dx = \text{line integration}$

$\iint dx dy = \text{surface integration}$

$\iiint dx dy dz = \text{volume integration}$

Gauss Divergence Theorem:

→ According to Gauss divergence theorem, the surface integration of a vector field ($\vec{A}$) over a close path is equal to volume integration of divergence of that field.

$$\oiint_S \vec{A} \cdot d\vec{s} = \iiint_V \vec{\nabla} \cdot \vec{A} dv$$

$dA = ds = \text{small surface area}$

$dv = \text{small volume element}$

Stoke's Theorem

→ According to Stoke's theorem, the line integration of a vector field $\vec{A}$ over a closed path is equal to surface integration of curl of that field.

$$\oint \vec{A} \cdot d\vec{l} = \iint_S \nabla \times \vec{A} \cdot d\vec{s}$$

$d\vec{l}$ = small line element

$d\vec{s}$ = small surface area

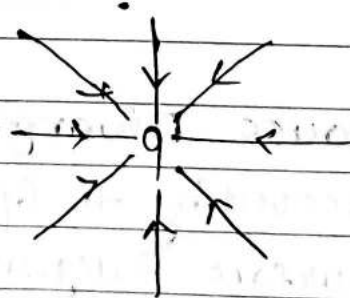
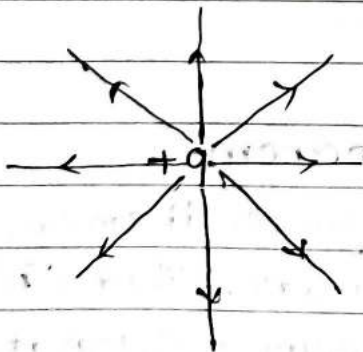
Electric field

→ The region around the source charge up to which the test charge Q can experience the force exerted by source charge is called electric field.

→ Electric field $\vec{E} = \lim_{Q \rightarrow 0} \frac{\vec{F}}{Q}$

$$\vec{E} = \frac{K q_1 q_2}{r^2}$$

Physical meaning of divergence



→ Divergence of electric field $\vec{E}$ tells us the flow of electric field lines inward / outward from of any surface S .

Divergence = outward / inward flow

$$\text{Divergence of Electric field } \vec{E} = \nabla \cdot \vec{E}$$

- Charges at rest produce electric field
- Charges in motion produce magnetic field.

$V = \text{Scalar}$

$$\vec{\nabla} V = \text{gradient of } V = \text{grad}(V)$$

Gauss law in electrostatic (in Integral form)

- The electric flux through a closed surface S is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed in it.

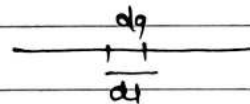
$$\text{Electric flux} = \boxed{\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}} \quad \begin{array}{l} \text{Integral form} \\ \text{of Gauss law} \end{array}$$

* Line Charge

- Line charge density λ

$$\lambda = \frac{\text{Charge}}{\text{length}} = \frac{dq}{dl}$$

$$dq = \lambda dl \Rightarrow \boxed{q = \int \lambda dl}$$

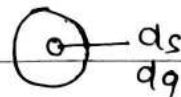


* Surface Charge

surface charge density σ

$$\sigma = \frac{\text{Charge}}{\text{Area}} = \frac{dq}{ds}$$

$$dq = \sigma ds \Rightarrow \boxed{q = \iint \sigma ds}$$

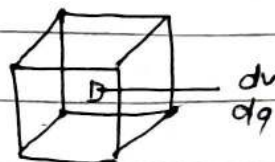


* Volume Charge

volume charge density ρ

$$\rho = \frac{\text{Charge}}{\text{volume}} = \frac{dq}{dv}$$

$$\Rightarrow dq = \rho dv \Rightarrow \boxed{q = \iiint \rho dv}$$



Divergence of electric field $\vec{E}$ $(\vec{\nabla} \cdot \vec{E})$

→ The integral form of Gauss's law.

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} \quad \text{--- (1)}$$

using Gauss Divergence Theorem, L.H.S becomes

$$\iiint_V \vec{\nabla} \cdot \vec{E} \, dv = \frac{1}{\epsilon_0} \iiint_V \rho \, dv$$

$$\iiint_V \vec{\nabla} \cdot \vec{E} \, dv - \iiint_V \frac{\rho}{\epsilon_0} \, dv = 0 \quad \left[\because \rho = \text{volume charge density} \right]$$

$$\iiint_V \left(\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right) dv = 0$$

Since $dv \neq 0$

$$\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0} = 0$$

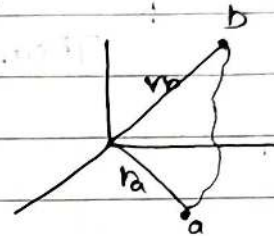
$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}} = \text{div } \vec{E}$$

This is differential form of Gauss law.

Curl of Electric field $\vec{E}$ ($\nabla \times \vec{E}$)

→ The electric field produced ^{due} to a charge q is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$



Take line integral both sides

$$\int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dl = \frac{q}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dl$$

$$= \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_a^b = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

for closed path

$$r_a = r_b$$

$$\oint \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_a} - \frac{1}{r_a} \right]$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$

Using Stoke's Theorem L.H.S

$$\iint_S (\nabla \times \vec{E}) \cdot d\vec{s} = 0$$

$$d\vec{s} \neq 0$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = 0}$$

$\times$ = cross product

$\oint$ = circulation

∇ - curl

→ It means that electric field lines do not form circular loops.

'curl' means = Circulation

~~then~~ Since,

$$\vec{\nabla} \times \vec{E} = 0$$

$$\text{Then, } \boxed{\vec{E} = -\vec{\nabla} v}$$

, v = scalar potential

$$\vec{E} = -\text{grad}(v)$$

$$\vec{E} = -\text{gradient of } v$$

V.I

4 marks

Laplace Theorem and Poisson Theorem.

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{--- (1)}$$

$$\vec{E} = -\vec{\nabla} v \quad \text{--- (2)}$$

Put (2) in (1)

$$\vec{\nabla} \cdot (-\vec{\nabla} v) = \frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2 v = -\frac{\rho}{\epsilon_0}}$$

Poisson
Theorem

(3)

ρ = volume charge density

$\vec{E}$ = electric field

$\vec{\nabla}$ = del operator

v = scalar ~~prod~~
potential

In charge free region

charge density $\Rightarrow \rho = 0$

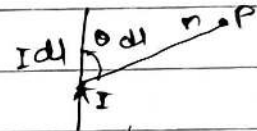
Put $\rho = 0$ in eq (3)

$$\boxed{\nabla^2 v = 0} \text{ --- Laplace Theorem}$$

Magnetic field

→ The region around a current carrying conductor where the magnetic effects can be experienced is called magnetic field.

Biot-Savart law



→ According to Biot-Savart law, dB = Small magnetic field.

$$dB \propto Idl$$

$$\propto \sin \theta$$

$$\propto \frac{1}{r^2}$$

$$dB \propto \frac{Idl \sin \theta}{r^2}$$

$$dB = k \frac{Idl \sin \theta}{r^2}$$

$$\therefore k = \frac{\mu_0}{4\pi}$$

$$dB = \frac{\mu_0 Idl \sin \theta}{4\pi r^2} \hat{r}$$

$$\hat{r} = \frac{\vec{r}}{r}$$

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

$$\boxed{\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \vec{r}}{r^3}}$$

Ampere's Circuital law

→ Integral form of Ampere's Circuital law.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

 μ_0 = permeability of free space. $d\vec{l}$ = line element. $d\vec{s}$ or $d\vec{A}$ = surface area dv = volume of element.

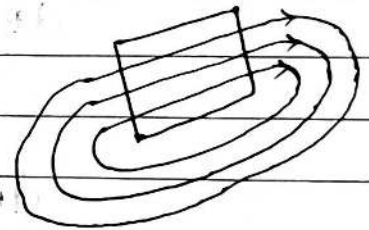
→ The line integration of magnetic field over a closed path is equal to μ_0 times the current flowing through it.

Divergence of Magnetic field $\vec{B}$ ($\nabla \cdot \vec{B}$)

→ The Gauss law in Magnetostatics

$$\oiint \vec{B} \cdot d\vec{s} = \text{Magnetic flux}$$

$$\oiint \vec{B} \cdot d\vec{s} = 0$$



→ Since magnetic lines of force form circular loops, so, the number of magnetic lines leaving the surface is equal to number of magnetic lines entering a surface. So, total magnetic flux

$$\oiint \vec{B} \cdot d\vec{s} = 0$$

Using Gauss Divergence Theorem.

$$\iiint \nabla \cdot \vec{B} \, dv = 0$$

Since, $du \neq 0$

$$\Rightarrow \boxed{\nabla \cdot \vec{B} = 0}$$

* Physical meaning

- magnetic monopoles do not exist.
- Magnetic lines of force always form circular loops.

Curl of magnetic field $\vec{B}$ ($\nabla \times \vec{B}$)

→ The Ampere Circuital law in integral form is

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I \quad \text{--- (1)}$$

$I = \text{current}$

using Stoke's Theorem

$$\oint_L \vec{B} \cdot d\vec{l} = \iint_S \nabla \times \vec{B} \cdot d\vec{s}$$

Put this value in L.H.S of eq (1)

$$\iint_S \nabla \times \vec{B} \cdot d\vec{s} = \mu_0 I \quad \text{--- (2)}$$

We know that the current density $\vec{J}$ for small current dI through a small surface area element ds is current density ($\vec{J}$) =

$$\vec{J} = \frac{\text{current}}{\text{Area}} = \frac{dI}{ds}$$

$$dI = \vec{J} \cdot d\vec{s}$$

$$I = \iint_S \vec{J} \cdot d\vec{s}$$

Put the value in R.H.S of eq (2)

$$\iint_S \vec{\nabla} \times \vec{B} \cdot d\vec{s} = \mu_0 \iint_S \vec{J} \cdot d\vec{s} = \iint_S \mu_0 \vec{J} \cdot d\vec{s}$$

$$\iint_S (\vec{\nabla} \times \vec{B} - \mu_0 \vec{J}) \cdot d\vec{s} = 0$$

Since, $d\vec{s} = 0$ small area element.

$$\vec{\nabla} \times \vec{B} - \mu_0 \vec{J} = 0$$

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}} \rightarrow \text{curl of } \vec{B}$$

↳ This is the differential form of
Ampere Circuital law.

V.V.V.
6marks

→ Calculation of vector potential ($\vec{A}$) for a given magnetic field by using Stoke's Theorem.

→ we know that from Biot - Savart laws, magnetic field due to a current loop is given by.

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \frac{d\vec{u} \times \vec{r}}{r^3} \quad \text{--- (1)}$$

we know that $\frac{\vec{r}}{r^3} = -\vec{\nabla} \left(\frac{1}{r} \right)$ --- To remember

$$\vec{B} = -\frac{\mu_0}{4\pi} \oint_L d\vec{u} \times \left[\vec{\nabla} \left(\frac{1}{r} \right) \right]$$

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} \quad \text{--- (2)} \quad [\because \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}]$$

$\therefore \frac{1}{r} = \text{scalar}$

$\therefore d\vec{u} = \text{vector}$

$$\vec{\nabla} \times (p\vec{a}) = p \vec{\nabla} \times \vec{a} - \vec{a} \times \underline{\vec{\nabla} p}$$

$$\vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) = \frac{1}{r} \vec{\nabla} \times d\vec{u} - d\vec{u} \times \underline{\vec{\nabla} \left(\frac{1}{r} \right)}$$

$$\vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) = \frac{1}{r} \vec{\nabla} \times d\vec{u} + \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} \quad [\because \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}]$$

$$\therefore \vec{\nabla} \left(\frac{1}{r} \right) \times d\vec{u} = \vec{\nabla} \times \left(\frac{1}{r} d\vec{u} \right) - \frac{1}{r} \vec{\nabla} \times d\vec{u}$$

Put this value in eq (2)

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \times \left(\frac{1}{r} d\vec{l} \right) - \frac{1}{r} \underbrace{\vec{\nabla} \times d\vec{l}}_0$$

$$\therefore \boxed{\vec{\nabla} \times d\vec{l} = 0}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \vec{\nabla} \times \left(\frac{1}{r} d\vec{l} \right)$$

$$\vec{B} = \oint_L \vec{\nabla} \times \left(\frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l} \right)$$

$$\vec{B} = \vec{\nabla} \times \oint \frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l} \quad \text{----- (3)}$$

Since,

$$\boxed{\vec{B} = \vec{\nabla} \times \vec{A}} \quad \text{----- (4)}$$

Comparing (3) and (4)

$$\vec{A} = \oint \frac{\mu_0}{4\pi} \frac{1}{r} d\vec{l}$$

$$\boxed{\begin{array}{l} \vec{\nabla} \cdot \vec{B} = 0 \\ \hookrightarrow \text{then } \vec{B} = \vec{\nabla} \times \vec{A} \end{array}}$$

Ques. Prove that $\vec{\nabla} \times \vec{d} = 0$

$$\vec{d} = \hat{i} dx + \hat{j} dy + \hat{k} dz$$

$$\vec{\nabla} = \hat{i} \frac{d}{dx} + \hat{j} \frac{d}{dy} + \hat{k} \frac{d}{dz}$$

$$\vec{\nabla} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ dx & dy & dz \end{vmatrix}$$

$$= \hat{i} \left(\frac{dz}{dy} - \frac{dy}{dz} \right) + \hat{j} \left(\frac{dx}{dz} - \frac{dz}{dx} \right) + \hat{k} \left(\frac{dy}{dx} - \frac{dx}{dy} \right)$$

$$= \hat{i} (0 - 0) + \hat{j} (0 - 0) + \hat{k} (0 - 0)$$

$$= (\hat{i} \times 0) + (\hat{j} \times 0) + (\hat{k} \times 0)$$

$$= 0 + 0 + 0$$

$$= 0$$

Unit-2 part-B

Date: _____

Electromagnetic waves

Equation of continuity

Physical meaning;

$$\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0$$

$\vec{J}$ = current density

ρ = volume charge density.

This equation tells us about the conservation of charge.

current $= I$ = flow of charge

$$I = \frac{-dq}{dt} \quad \text{--- (1)}$$

We know that, if ρ is volume charge density, then current in volume V is

$$q = \iiint_V \rho \, dv \quad \text{--- (2)}$$

$$q = \iiint \rho \, dv$$

$$I = \iint_S \sigma \, ds$$

$$q = \int I \, dt$$

$$\text{current } \vec{J} = \frac{\text{current}}{\text{Area}} = \frac{dI}{ds}$$

$$dI = \vec{J} \cdot d\vec{s}$$

$$I = \oint_S \vec{J} \cdot d\vec{s} \quad \text{--- (3)}$$

Put values from (2) and (3) in (1)

$$\oint_S \vec{J} \cdot d\vec{s} = -\frac{q}{dt} \iiint_V \rho dv$$

using Gauss Divergence Theorem L.H.S becomes

$$\iiint_V \vec{\nabla} \cdot \vec{J} dv = -\iiint_V \frac{d\rho}{dt} dv$$

$$\iiint_V \vec{\nabla} \cdot \vec{J} dv + \iiint_V \frac{d\rho}{dt} dv = 0$$

$$\iiint_V \left(\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} \right) dv = 0$$

Since $dv \neq 0$

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0}$$

This is equation of continuity.

Unit 10
(6 mark)

Maxwell's Equation in Differential form are :

① $\vec{\nabla} \cdot \vec{D} = \rho$, $\vec{D} = \epsilon_0 \vec{E}$ = Gauss's law in electrostatic

②. $\vec{\nabla} \cdot \vec{B} = 0$ Gauss's law in Magnetostatic

$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$ Faraday's law of Electromagnetic Induction

$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$, $\vec{B} = \mu_0 \vec{H}$ Modified Ampere
Circular law

Proof of Maxwell's Equation differential form.

①. $\boxed{\vec{\nabla} \cdot \vec{D} = \rho} \rightarrow \text{To Prove}$

Proof: The Gauss's law in Electrostatic is

$$\oint \vec{E} \cdot d\vec{s} = \frac{\rho q}{\epsilon_0}$$

$$\oint (\epsilon_0 \vec{E}) \cdot d\vec{s} = \rho q$$

$$\oint \vec{D} \cdot d\vec{s} = \rho q$$

Using Gauss's Divergence theorem in L.H.S

$$\iiint \vec{\nabla} \cdot \vec{D} \, dv = \rho q$$

$$\text{Put } \rho = \iiint \rho \, dv$$

$$\iiint \vec{\nabla} \cdot \vec{D} \, dv = \iiint \rho \, dv$$

$$\iiint (\vec{\nabla} \cdot \vec{D} - \rho) \, dv = 0$$

$$dv \neq 0$$

$$\vec{\nabla} \cdot \vec{D} - \rho = 0$$

$$\boxed{\vec{\nabla} \cdot \vec{D} = \rho}$$

Proved

②. Maxwell second Equation.

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0} \rightarrow \text{Prove}$$

Proof: Gauss law in magnetostatic is

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

Using Gauss Divergence Theorem.

$$\iiint_V (\vec{\nabla} \cdot \vec{B}) \, dv = 0$$

$$dv \neq 0$$

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

Proved

Maxwell's third Equation

$$\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt} \longrightarrow \text{To prove}$$

Proof :-

According to Faraday's law of electromagnetic induction, the e.m.f is induced given by

$$e = - \frac{d\phi}{dt}, \quad \phi = \text{magnetic flux.} \quad \text{--- (1)}$$

$$\text{magnetic flux } \phi = \iint \vec{B} \cdot d\vec{s}$$

Put this value in eq (1)

$$e = - \frac{d}{dt} \iint \vec{B} \cdot d\vec{s}$$

$$e = - \iint \frac{d\vec{B}}{dt} \cdot d\vec{s} \quad \text{--- (2)}$$

The e.m.f is defined as the work done to take a unit positive charge around a closed loop.

$$e = \oint \vec{E} \cdot d\vec{l} \quad \text{--- (3)}$$

Put this value in L.H.S of eq (2)

$$\oint \vec{E} \cdot d\vec{l} = - \iint \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

using Stoke's Theorem in L.H.S

$$\iint_S \vec{\nabla} \times \vec{E} \, d\vec{s} = - \iint_S \frac{d\vec{B}}{dt} \, d\vec{s}$$

$$\iint_S \vec{\nabla} \times \vec{E} \, d\vec{s} + \iint_S \frac{d\vec{B}}{dt} \, d\vec{s} = 0$$

$$\iint_S \left(\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} \right) d\vec{s} = 0$$

since, $d\vec{s} \neq 0$

$$\vec{\nabla} \times \vec{E} + \frac{d\vec{B}}{dt} = 0$$

$$\boxed{\vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}}$$

Hence, Proved

Maxwell's fourth Equation

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt} \quad \longrightarrow \text{To prove}$$

Proof:-

for time independent currents, the differential form of Ampere's circuital law is

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\vec{B} = \mu_0 \vec{H}$$

$$\boxed{\frac{\vec{B}}{\mu_0} = \vec{H}}$$

$$\vec{\nabla} \times \frac{\vec{B}}{\mu_0} = \vec{J}$$

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J}}$$

According to modified Ampere's circuital law,
for time dependent currents, we have

$$\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}' \quad \text{--- (1)}$$

$$J' = \frac{dD}{dt}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{dD}{dt}$$

Proved

According to Equation of continuity

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{d\rho}{dt} = 0}$$

marks

ρ = charge density

$\vec{J}$ = current density

$$\vec{\nabla} \cdot \vec{J} = -\frac{d\rho}{dt} \quad \text{--- (2)}$$

When divergence of eq (1) both sides

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot (\vec{J} + \vec{J}') \quad \text{--- (3)}$$

Since,

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0$$

$$\vec{\nabla} \cdot (\vec{J} + \vec{J}') = 0$$

$$\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}' = 0$$

using eq (2)

$$-\frac{d\rho}{dt} + \vec{\nabla} \cdot \vec{J}' = 0$$

$\therefore J' = \text{Time varying current density}$

$$\vec{\nabla} \cdot \vec{J} = \frac{d\rho}{dt} \quad \text{--- (4)}$$

Now,

Maxwell first equation is

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Put $\rho = \vec{\nabla} \cdot \vec{D}$ in R.H.S of eq (4)

$$\vec{\nabla} \cdot \vec{J} = \frac{d\rho}{dt} = \frac{d(\vec{\nabla} \cdot \vec{D})}{dt}$$

$$\vec{\nabla} \cdot \vec{J} = \vec{\nabla} \cdot \left(\frac{d\vec{D}}{dt} \right)$$

$$\Rightarrow \vec{J} = \frac{d\vec{D}}{dt} \quad \rightarrow \text{Put this value in eq (1)}$$

$$\boxed{\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}}$$

Hence, Proved.

Maxwell's Equations in differential form:

(i) $\vec{\nabla} \cdot \vec{D} = \rho$ $\vec{D} = \epsilon_0 \vec{E} \rightarrow$ Gauss law in electrostatic.

(ii) $\vec{\nabla} \cdot \vec{B} = 0$ $\rightarrow$ Gauss law in magnetostatic

(iii) $\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \rightarrow$ Faraday's law of electromagnetic induction.

(iv) $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$, $\vec{B} = \mu_0 \vec{H} \rightarrow$ Modified form of Ampere circuital law.

In free space or vacuum

$$\rho = 0, \quad J = 0$$

$$(i) \quad \vec{\nabla} \cdot \vec{D} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

$$\vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt}$$

In conducting medium.

$$\vec{J} = \sigma \vec{E} \quad \sigma = \text{conductivity.}$$

$$\vec{J} \neq 0$$

6 marks

Maxwell's Equation in Integral form:

(i) Gauss law in electrostatics

$$\text{To Prove} - \oint \vec{D} \cdot d\vec{s} = q, \quad \vec{D} = \epsilon_0 \vec{E}$$

Proof:-

The differential form of Gauss's law or Maxwell's first equation is

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Take volume integration both sides.

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \iiint_V \rho \, dv \quad \text{--- (1)}$$

using Gauss divergence theorem

$$\iiint_V \vec{\nabla} \cdot \vec{D} \, dv = \oiint_S \vec{D} \cdot d\vec{s}$$

$$q = \iiint_V \rho \, dv$$

Put this value in eq (1)

$$\boxed{\oiint_S \vec{D} \cdot d\vec{s} = q} \quad \text{Integral form of Gauss law in electro static.}$$

(EE) Gauss law in Magnetostatic

$$\boxed{\oiint_S \vec{B} \cdot d\vec{s} = 0} \quad \text{— To prove}$$

Proof:-

Maxwell's second equation in differential form is

$$\vec{\nabla} \cdot \vec{B} = 0$$

Take volume integration both sides.

$$\iiint_V \vec{\nabla} \cdot \vec{B} \, dv = 0$$

using Gauss divergence theorem.

$$\iiint_V \vec{\nabla} \cdot \vec{B} \, dv = \oiint_S \vec{B} \cdot d\vec{s}$$

$$\boxed{\oiint_S \vec{B} \cdot d\vec{s} = 0} \quad \text{proved}$$

(iii) Maxwell's Third Equation in Integral form.

To Prove -
$$\oint_L \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{s}$$

Proof:

Maxwell's third equation in differential form

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

Take surface integration both sides.

$$\iint_S \vec{\nabla} \times \vec{E} \cdot d\vec{s} = -\iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

using Stoke's theorem in L.H.S

$$\oint_L \vec{E} \cdot d\vec{r} = -\iint_S \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\boxed{\oint_L \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{s}}$$

Proved

(iv) Maxwell's fourth Equation in Integral form

$$\oint_L \vec{H} \cdot d\vec{r} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \cdot d\vec{s}$$

Proof:

Consider Maxwell's 4th Equation in differential form

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

Take surface integration both sides

$$\iint_S \vec{\nabla} \times \vec{H} \, d\vec{s} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) d\vec{s}$$

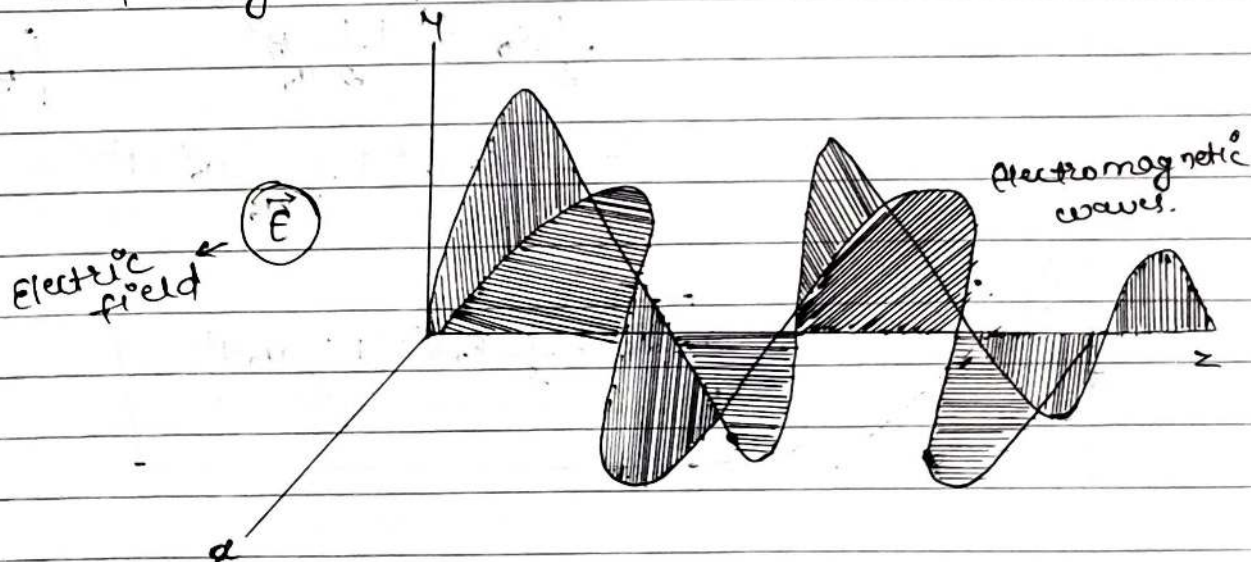
using Stoke's Theorem in L.H.S

$$\oint_L \vec{H} \cdot d\vec{s} = \iint_S \left(\vec{J} + \frac{d\vec{D}}{dt} \right) d\vec{s}$$

Hence Proved.

Electromagnetic waves (EM waves) Electromagnetic Energy

→ Electromagnetic wave are the waves that are created as a result of vibrations between electric and magnetic field.



102
2 marks

Energy stored in electromagnetic waves or
electromagnetic energy.

Energy stored in electric field

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

Energy stored in magnetic field

$$U_B = \frac{1}{2} \mu_0 B^2$$

Total Energy stored in electromagnetic waves

$$U_{EM} = U_E + U_B$$

$$= \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 B^2$$

$$\boxed{\vec{B} = \mu_0 \vec{H}}$$

$$= \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0^2 H^2$$

$$\boxed{U_E = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 H^2}$$

6 Marks
Agile assessment
mon 20/09/20

Poynting's Theorem

The Poynting theorem says that the work done on the charges by electromagnetic waves is equal to decrease in the energy stored in the field, and minus of energy that thrown out to the surface.

Proof:

consider Maxwell's fourth equation in differential form.

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt} \quad \text{--- (1)}$$

Take dot product with $\vec{E}$ both sides

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{E} \cdot \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \quad \text{--- (2)}$$

~~$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{A} \times \vec{C}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$~~

Using standard vector identity

~~$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H})$$~~

To Remember :- $\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H})$

~~$$\Rightarrow \vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H})$$~~

Put this value in eq (2)

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \left(\vec{J} + \frac{d\vec{D}}{dt} \right) \quad \text{--- (3)}$$

Since $\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$

$$\boxed{\vec{D} = \epsilon_0 \vec{E}}$$

Put this eq in (3)

$$\vec{H} \cdot \left(-\frac{d\vec{B}}{dt} \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E})$$

$$\vec{B} = \mu_0 \vec{H} \Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0}$$

$$\frac{\vec{B}}{\mu_0} \cdot \left(-\frac{d\vec{B}}{dt} \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E})$$

$$-\frac{\vec{B}}{\mu_0} \cdot \frac{d\vec{B}}{dt} - \vec{E} \cdot \frac{d}{dt} (\epsilon_0 \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$-\frac{d}{dt} \left(\frac{1}{2\mu_0} \vec{B}^2 \right) - \frac{d}{dt} \left(\frac{1}{2} \epsilon_0 \vec{E}^2 \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$\therefore \frac{d}{dt} \left(\frac{1}{2\mu_0} B^2 \right) = \frac{1}{\mu_0} B \frac{dB}{dt} = \frac{1}{\mu_0} B d\vec{B}$$

$$\frac{d}{dt} \left(\frac{1}{2} \epsilon_0 E^2 \right) = \frac{1}{2} \epsilon_0 \frac{dE^2}{dt} = \epsilon_0 E dE$$

$$-\frac{d}{dt} \left[\frac{1}{2\mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2 \right] - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J}$$

$$\iiint_V \vec{E} \cdot \vec{J} dV = -\frac{d}{dt} \iiint_V \left(\frac{1}{2\mu_0} B^2 + \frac{1}{2} \epsilon_0 E^2 \right) dV - \iiint_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV \quad (4)$$

using Gauss divergence theorem in last term of R.H.S

$$\iiint_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dv = \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

Put this equation in (4) :

$$\iiint_V \vec{E} \cdot \vec{J} dv = -\frac{d}{dt} \iiint_V \left(\frac{1}{2\mu_0} B^2 + \frac{1}{2\epsilon_0} E^2 \right) dv - \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

The left hand side represents the work done on the charges by electromagnetic field. The first term in R.H.S represents the negative time derivative energy stored in electromagnetic waves and the second term represents the energy carried by electromagnetic waves during its outward flow.

UoVof

1 or 2 Marks

Poynting vector ($\vec{S}$)

$$\vec{S} = \vec{E} \times \vec{H}$$

$\vec{E}$ = Electric field

$\vec{H}$ = Magnetic field, $\vec{B} = \mu_0 \vec{H}$

→ Poynting vectors tells us the rate of flow of electromagnetic waves per unit area per unit time.

The wave Equation of Electromagnetic in free space.

→ In free space Maxwell's four equation are:

$$(i) \vec{\nabla} \cdot \vec{D} = 0 \text{ --- (1)}$$

$$\rho = 0$$

$$(ii) \vec{\nabla} \cdot \vec{B} = 0 \text{ --- (2)}$$

$$J = 0$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{B} = \mu_0 \vec{H}$$

$$(iii) \vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \quad (3)$$

$$(iv) \vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt} \quad (4)$$

Wave Eqn for electric field

Maxwell's third equation =

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$$

Take curl both sides

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left(-\frac{d\vec{B}}{dt} \right) \quad (5)$$

Using vector triple product

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - (\vec{A} \cdot \vec{B})\vec{C}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - (\vec{\nabla} \cdot \vec{\nabla})\vec{E} \quad (6)$$

Since,

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot (\epsilon_0 \vec{E}) = 0$$

$$\therefore \vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

Put these values in eqn (5) (6)

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = 0 - \nabla^2 \vec{E}$$

$$= -\nabla^2 \vec{E}$$

So,

eqn (5) becomes

$$-\nabla^2 E = \vec{\nabla} \times \left(-\frac{d\vec{B}}{dt} \right)$$

$$= -\frac{d}{dt} (\vec{\nabla} \times \vec{B})$$

$$\text{Put } \vec{B} = \mu_0 \vec{H}$$

$$-\nabla^2 \vec{E} = -\frac{d}{dt} (\vec{\nabla} \times \mu_0 \vec{H})$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d}{dt} (\vec{\nabla} \times \vec{H})$$

$$\text{Put } \vec{\nabla} \times \vec{H} = \frac{d\vec{D}}{dt} \quad (\text{Maxwell's fourth Eqn in free space})$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d}{dt} \left(\frac{d\vec{D}}{dt} \right) \quad \begin{matrix} \vec{B} = \mu_0 \vec{H} \\ \vec{D} = \epsilon_0 \vec{E} \end{matrix}$$

$$\nabla^2 \vec{E} = \mu_0 \frac{d^2}{dt^2} (\epsilon_0 \vec{E})$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2}$$

Since,

$$\text{Speed of light } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\Rightarrow c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\Rightarrow \frac{1}{c^2} = \frac{1}{\mu_0 \epsilon_0}$$

Put this value in eq (7)

$$\nabla^2 E = \frac{1}{c^2} \frac{d^2 E}{dt^2}$$

This is EM waves equation for electric field.

Similarly, for magnetic field wave equation is

$$\nabla^2 H = \frac{1}{c^2} \frac{d^2 H}{dt^2}$$

Plane electromagnetic waves in free space/vacuum.

(Prove that electromagnetic waves are transverse in nature)

Consider a plane electromagnetic wave travelling along z-axis only.

for such a wave electric field and magnetic vectors are represented by

$$\vec{E} = \vec{E}_0 e^{i(\omega t - kz)} \quad \text{--- (1)}$$

$$\vec{H} = \vec{H}_0 e^{i(\omega t - kz)} \quad \text{--- (2)}$$

$$\vec{\nabla} = \hat{i} \frac{d}{dx} + \hat{j} \frac{d}{dy} + \hat{k} \frac{d}{dz}$$

for z direction,

$$\vec{\nabla} = \hat{k} \frac{d}{dz} \quad \text{--- (3)}$$

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla}$$

$$\nabla^2 = \frac{d^2}{dz^2} \quad \text{--- (4)}$$

Now, the wave eq.ⁿ for electromagnetic waves is

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{d^2 \vec{E}}{dt^2} \quad \text{--- (5)}$$

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{d^2 \vec{H}}{dt^2} \quad \text{--- (6)}$$

Take derivative of (1) w.r.t 'z'

$$\frac{d\vec{E}}{dz} = -ik\epsilon_0 e^{i(\omega t - kz)}$$

$$\frac{d\vec{E}}{dz} = -ik\vec{E}$$

$$\boxed{\frac{d}{dz} = -ik} \quad \text{--- (7)}$$

Now, take derivative of (1) w.r.t 't'

$$\frac{d\vec{E}}{dt} = i\omega \epsilon_0 e^{i(\omega t - kz)}$$

$$\frac{d\vec{E}}{dt} = i\omega \vec{E}$$

$$\boxed{\frac{d}{dt} = i\omega} \quad \text{--- (8)}$$

Now Maxwell's first and second eqⁿ in freespace ($\rho=0$) are

$$\vec{\nabla} \cdot \vec{D} = 0$$

$$\left(\hat{k} \frac{d}{dz} \right) \cdot (\epsilon_0 \vec{E}) = 0$$

$$\Rightarrow \hat{k} \frac{d}{dz} \cdot \vec{E} = 0$$

using eq (7)

$$\Rightarrow \hat{k} (-i k) \cdot \vec{E} = 0$$

$$\Rightarrow -i k (\hat{k} \cdot \vec{E}) = 0$$

$$\Rightarrow \boxed{\hat{k} \cdot \vec{E} = 0}$$

$\hat{k}$ = direction of electromagnetic

Hence EM wave and electric field $\vec{E}$ for $\perp$ to each other.

Again,

for Maxwell second eqⁿ.

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot (\mu_0 \vec{H}) = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{H} = 0$$

$$\hat{k} \frac{d}{dz} \cdot \vec{H} = 0$$

using eq (8)

$$\hat{k} (-i k) \cdot \vec{H} = 0$$

$$-i k (\hat{k} \cdot \vec{H}) = 0$$

$$\Rightarrow \boxed{\hat{k} \cdot \vec{H} = 0}$$

EM waves and Magnetic field $\vec{B}$ are $\perp$ to each other.

UNIT-3

Date : _____

WAVE NATURE OF PARTICLES AND SCHRÖDINGER EQUATION

Macroscopic world
Classical Mechanics

$$v = u + at$$

$$v^2 - u^2 = 2as$$

$$s = ut + \frac{1}{2}at^2$$

Microscopic world
electron, proton, neutron.

Dual nature of light

→ In some cases light behaves as wave in other cases it behaves as a particles.

* Wave Nature

(i) Interference

(ii) Diffraction

(iii) Polarization

* Particles Nature

(i) Photoelectric effect

(ii) Compton scattering

(iii) Pair Production

de-Broglie Hypothesis

→ According to de Broglie Hypothesis every moving particles having mass 'm' characteristics of both wave and particles.

→ The wavelength associated with particles with mass 'm' is

$$\lambda_{\text{de brog}} = \frac{h}{mv}$$

h = Planck's constant = 6.634×10^{-34} Js

m = Mass of particle

v = Velocity of particles

wave function $\psi(x, t) \rightarrow 1-D$

$\psi(x, y, t) \rightarrow 2-D$

$\psi(x, y, z, t) \rightarrow 3-D$

→ wave function gives wth the state of a microscopic particle of mass 'm' at point 'x' at any time 't'.

The wave function of particles having mass 'm' is represented by

$$\psi(x, t) = A e^{i(kx - \omega t)} \quad (1)$$

$k = \frac{2\pi}{\lambda} \rightarrow$ wave vector

$\omega = 2\pi \nu$ angular frequency

$\nu =$ frequency

$$\lambda = \frac{h}{mv} = \frac{h}{p} \quad p = \text{linear momentum}$$

$$E = h\nu, \quad \nu = \frac{E}{h}$$

$$[\because \hbar = \frac{h}{2\pi}]$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/p} = \frac{2\pi p}{h} = \frac{p}{h/2\pi} = \frac{p}{\hbar}$$

$h =$ Planck's constant

$\hbar =$ Reduced value of Planck's constant.

$$\omega = 2\pi\nu = 2\pi \frac{E}{h} = \frac{E}{h/2\pi} = \frac{E}{\hbar}$$

Put these value in eq (1)

$$\psi(x, t) = A e^{i\left(\frac{p}{\hbar}x - \frac{Et}{\hbar}\right)}$$

$$\psi(x, t) = A e^{\frac{i}{\hbar}(px - Et)}$$

$$\psi(x, t) = A e^{i(kx - \omega t)}$$

Remember.

Uncertainty Relation

Heisenberg Uncertainty Relation.

→ According to Uncertainty Relation it is impossible to simultaneously find the position and momentum of a microscopic particles with 100% accuracy.

Δx = Error or uncertainty in the measurement of position of particles.

Δp = momentum of particles.

then

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

If $\Delta x = 0 \rightarrow$ Error in position = 0

$$\Delta p \geq \frac{\hbar}{2 \Delta x}$$

$$\Delta p \geq \frac{1}{0} = \infty$$

If $\Delta p = 0$

$$\Delta x = \frac{\hbar}{\Delta p} = \frac{\hbar}{0} = \infty$$

Born Interpretation

→ According to Born Interpretation the probability of existence of a particle at any point 'x' is

$$P = \psi^* \psi = |\psi|^2$$

$x = x_0$

In a region $a \leq x \leq b$, the probability of existence is given by

$$P = \int_a^b \psi^* \psi dx$$

$$\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

$$\psi^*(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

(i) Ques. find probability at point $x = \frac{L}{2}$

$$P = \psi^* \psi = \frac{2}{L} \sin^2 \frac{\pi x}{L}$$

$x = \frac{L}{2}$

$$\frac{2}{L} \sin^2 \frac{\pi \cdot \frac{L}{2}}{L} = \frac{2}{L} \sin^2 \frac{\pi}{2}$$

$$P \text{ at point } x = \frac{L}{2} = \frac{2}{L}$$

$$P = \int_0^{L/2} \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

Ques.

4 marks

Using Heisenberg Uncertainty Relation Prove that electron cannot exist inside the nucleus.

Proof: The Heisenberg Uncertainty Relation

$$\Delta x \Delta p \approx \frac{h}{2}$$

$$\hbar = \frac{h}{2\pi}$$

$$\Delta x \Delta p \approx \frac{h}{4\pi} \quad \text{--- (1)}$$

If electron is inside the nucleus

$$\Delta x = 10^{-15} \text{ m} \quad \text{--- size of nucleus.}$$

$$\Delta p = m \Delta v$$

$\therefore v = \text{velocity of electron}$

$$m = 9.1 \times 10^{-31} \text{ kg} = \text{mass of } e^-$$

Put these values in eq (1)

$$\Delta x \cdot m \Delta v = \frac{h}{4\pi}$$

$$\Delta v = \frac{h}{4\pi m \Delta x}$$

$$= \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 10^{-15}}$$

$$\Delta v \approx 10^{10} \text{ ms}^{-1}$$

which is larger than the speed of light is impossible. Hence electron cannot exist inside the nucleus.

$$\hat{E} = \text{Eota}$$

$$\hat{E}^* = \text{complex conjugate} = -\hat{E}$$

Born Interpretation.

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\theta = \frac{\pi x}{L}$$

$$\sin^2 \frac{\pi x}{L} = \frac{1 - \cos \frac{2\pi x}{L}}{2}$$

Put in (1)

$$P = \frac{2}{L} \int_0^{L/2} \frac{1 - \cos \frac{2\pi x}{L}}{2} dx$$

$$P = \frac{1}{L} \left[\int_0^{L/2} 1 dx - \int_0^{L/2} \cos \frac{2\pi x}{L} dx \right]$$

$$= \frac{1}{L} \left[\left[x \right]_0^{L/2} - \left[\frac{\sin \frac{2\pi x}{L}}{\frac{2\pi}{L}} \right]_0^{L/2} \right]$$

$$= \frac{1}{2} \left[\left(\frac{L}{2} - 0 \right) - \frac{L}{2\pi} \left(\sin \frac{2\pi}{K} \cdot \frac{K}{2} - \sin 0 \right) \right]$$

$$= \frac{1}{2} \times \frac{L}{2}$$

$$\boxed{P = \frac{1}{2}}$$

Expectation value (Average value)

$$\langle f(x) \rangle = \frac{\int_{-\infty}^{\infty} \psi^* f(x) \psi dx}{\int_{-\infty}^{\infty} \psi^* \psi dx}$$

$\langle x \rangle$ = Expectation or average value of position.

$\langle p \rangle$ = Expectation or average value of linear momentum.

(Marks)

Normalization of wave function

Condition to normalize a wave function is

$$\int_{-\infty}^{\infty} \psi^* \psi dx = 1$$

If ψ is normalized

Ques. Consider a wave unnormalized wave function

$$\psi(x) = A \sin \frac{\pi x}{L}; 0 \leq x \leq L$$

$\therefore A$ = normalized constant

Normalized this wave function.

Solⁿ:- The condition to normalized a wave function

$$\int_{-\infty}^{\infty} \psi^* \psi dx = 1$$

$$\psi^* = \psi = A \sin \frac{\pi x}{L}$$

$$\int_0^L A \sin \frac{\pi x}{L} \cdot A \sin \frac{\pi x}{L} dx = 1$$

$$A^2 \int_0^L \sin^2 \frac{\pi x}{L} dx = 1$$

$$A^2 \int_0^L \frac{1 - \cos \frac{2\pi x}{L}}{2} dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{L}{2\pi} \sin \frac{2\pi x}{L} \right]_0^L = 1$$

$$\frac{A^2}{2} \left[(L-0) - 0 \right] = 1$$

$$A^2 \frac{L}{2} = 1$$

$$A^2 = \frac{2}{L}$$

$$A = \sqrt{\frac{2}{L}}$$

Put $A = \sqrt{\frac{2}{L}}$ is given wave function.

Hence, the normalized wave function is

$$\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

U.V.I

6 Marks

Time dependent Schrodinger Equation

Consider a particle of mass 'm' represented by wave function.

$$\psi(x, t) = A e^{\frac{i}{\hbar}(Et - px)} \quad \text{--- (1)}$$

E = Energy

p = linear momentum

x = position

t = time

Total energy of particle

$$E = K.E + P.E$$

$$E = \frac{1}{2} m v^2 + V(x)$$

$$E = \frac{1}{2} \frac{m^2 v^2}{m} + V(x)$$

$$E = \frac{p^2}{2m} + V(x)$$

Apply wave function ψ both sides

$$E\psi = \frac{p^2}{2m}\psi + V(x)\psi \quad \text{--- (2)}$$

Take derivative of (1) both sides w.r.t 't'

$$\frac{d\psi}{dt} = A e^{\frac{i}{\hbar}(Et - px)} \cdot \frac{-iE}{\hbar}$$

$$\frac{d\psi}{dt} = \frac{-iE}{\hbar} A e^{\frac{i}{\hbar}(Et - px)}$$

$$\frac{d\psi}{dt} = \frac{-iE}{\hbar} \psi$$

$$\boxed{\frac{\hbar}{-i} \frac{d\psi}{dt} = E\psi} \quad \text{--- (3)}$$

Take derivative of (1) both sides w.r.t 'x' two times.

$$\frac{d\psi}{dx} = A e^{-\frac{i}{\hbar}(Et - px)}$$

$$= A e^{-\frac{i}{\hbar}(Et - px)} \cdot -\frac{i}{\hbar}(-p)$$

$$\frac{d}{dx} \left(\frac{d\psi}{dx} \right) = A \frac{ip}{\hbar} \frac{d}{dx} \left[e^{-\frac{i}{\hbar}(Et - px)} \right]$$

$$\frac{d^2\psi}{dx^2} = \frac{ip}{\hbar} A e^{-\frac{i}{\hbar}(Et - px)} \frac{ip}{\hbar}$$

$$\frac{d^2\psi}{dx^2} = \frac{i^2 p^2}{\hbar^2} A e^{-\frac{i}{\hbar}(Et - px)} \underset{\psi}{\quad}$$

$$\frac{d^2\psi}{dx^2} = -\frac{p^2\psi}{\hbar^2}$$

$$\boxed{p^2\psi = -\hbar^2 \frac{d^2\psi}{dx^2}} \quad \text{--- (4)}$$

$$\text{but } E\psi = \hbar \frac{d\psi}{dt} \quad \text{and } p^2\psi = -\hbar^2 \frac{d^2\psi}{dx^2}$$

from (3) and (4) in eq (2)

$$\frac{\hbar}{-i} \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi$$

multiple and divide by i

$$\frac{i\hbar}{-i^2} \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi$$

$$\boxed{i\hbar \frac{d\psi}{dt} = -\hbar^2 \frac{d^2\psi}{dx^2} + V(x)\psi}$$

This is time-dependent schrodinger eqn in 1-D

for free particle $V(x) = 0$

$$i\hbar \frac{d\psi}{dt} = -\hbar^2 \frac{d^2\psi}{dx^2}$$

Time - Independent schrodinger Eqⁿ.

Time-dependent schrodinger equation is

$$-\hbar^2 \frac{d^2\psi}{dx^2} + V(x)\psi(x,t) = i\hbar \frac{d\psi}{dt} \quad \text{--- (1)}$$

let the time dependent wave function

$\psi(x,t)$ is written as

$$\psi(x,t) = \phi(x) e^{-\frac{iEt}{\hbar}} \quad \text{--- (2)}$$

Take derivative of (2) both sides w.r.t 'x' two times

$$\frac{d\psi}{dx} = \frac{d\phi(x)}{dx} e^{-\frac{iEt}{\hbar}}$$

$$\boxed{\frac{d^2\psi}{dx^2} = \frac{d^2\phi}{dx^2} e^{-\frac{iEt}{\hbar}}} \quad \text{--- (3)}$$

Take derivative of (2) both sides w.r.t 't'

$$\frac{d\psi}{dt} = \phi(x) e^{-\frac{iEt}{\hbar}} \cdot \left(-\frac{iE}{\hbar}\right) \quad \text{--- (4)}$$

Put values of $\frac{d^2\psi}{dx^2}$ and $\frac{d\psi}{dt}$ and ψ

from (3) and (4) in eq (1)

$$-\frac{\hbar^2}{2m} \left(\frac{d^2\phi}{dx^2} e^{-\frac{iEt}{\hbar}} \right) + V(x) \left[\phi(x) e^{-\frac{iEt}{\hbar}} \right] = i\hbar \left[\phi(x) e^{-\frac{iEt}{\hbar}} \left(\frac{-iE}{\hbar} \right) \right]$$

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2\phi(x)}{dx^2} + V(x)\phi(x) = E\phi(x)}$$

Time-independent schrodinger eqⁿ in 1-D for particle $[V(x) \approx 0]$

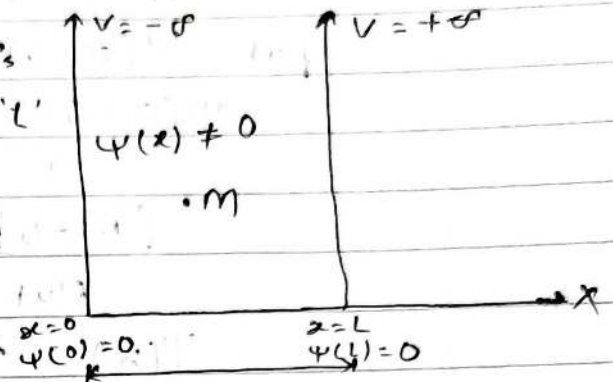
$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2\phi}{dx^2} = E\phi}$$

UoV.1
6 marks

Particle in a 1-Dimensional Box

m = mass of particle which is inside the box of length ' L '

particle is free inside the box
potential energy $V=0$



$\psi(x) \rightarrow$ represent state of particle

L = length of Box

$$V = 0 \quad 0 < x < L$$

$$V = +\infty \quad x \geq L, x \leq 0$$

The time-independent equation for a free particle in

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

Multiply both sides by $-\frac{2m}{\hbar^2}$

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2} \psi$$

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0$$

$$\text{Put } \frac{2mE}{\hbar^2} = k^2 \quad \text{--- (1)}$$

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{--- (2)}$$

The solution of eqn (2) is

$$\psi(x) = A \sin kx + B \cos kx \quad \text{--- (3)}$$

(i) Boundary condition of $x=0$:

at $x=0$, $\psi(x=0) = 0$

put $\psi(x=0) = 0$ and $x=0$ in --- (3)

$$\psi(x=0) = A \sin 0 + B \cos 0$$

$$0 = 0 + B$$

$$\boxed{B = 0}$$

Put $B = 0$ in eqⁿ (3)

$$\psi(x) = A \sin kx \text{ --- (4)}$$

(ii) Second Boundary condition at $x=L$, $\psi(x=L) = 0$

put this value in (4)

$$\psi(x=L) = 0 = A \sin kL$$

$$A \sin kL = 0 = \sin \pi$$

$\therefore n = 1, 2, 3, 4, \dots$

$$\Rightarrow kL = n\pi$$

$$\boxed{k = \frac{n\pi}{L}}$$

put this value in (1)

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2}$$

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}}$$

$n = 1, 2, 3, 4, \dots$

$$n_1 = E_1 = \frac{\pi^2\hbar^2}{2mL^2} \text{ is ground state energy}$$

$$n_2 = E_2 = 4 \frac{\pi^2\hbar^2}{2mL^2} = 4E_1 \text{ is energy of first excited state}$$

$$n_3 = E_3 = 9 \frac{\pi^2 \hbar^2}{2mL^2} \rightarrow E_1$$

= $9E_1$ = second energy of excited state.

$$n=4 \quad \text{-----} \quad 16E_1$$

$$n=3 \quad \text{-----} \quad 9E_1$$

$$n=2 \quad \text{-----} \quad 4E_1$$

$$n=1 \quad \text{-----} \quad E_1$$

To find wave function of particle

$$\psi(x) = A \sin kx$$

$$\text{put } k = \frac{n\pi}{L}$$

$$\psi(x) = A \sin \frac{n\pi x}{L} \quad \text{--- (5)}$$

Here, A = normalization constant

The condition to normalize of wave function is

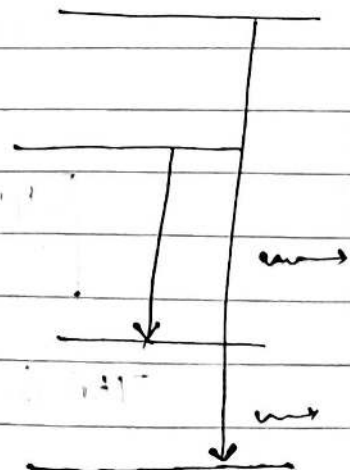
$$\int_{-\infty}^{+\infty} \psi^* \psi dx = 1$$

$$\psi^*(x) = A \sin \frac{n\pi x}{L}$$

$$\int_0^L A \sin \frac{n\pi x}{L} dx = 1$$

$$A^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = 1$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$



$$\sin \frac{2n\pi x}{L} = \frac{1 - \cos \frac{2n\pi x}{L}}{2}$$

$$A^2 \int_0^L \frac{1 - \cos \frac{2n\pi x}{L}}{2} dx = 1$$

$$\frac{A^2}{2} \int_0^L \left(1 - \cos \frac{2n\pi x}{L} \right) dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{\sin \frac{2n\pi x}{L}}{\frac{2n\pi}{L}} \right]_0^L$$

$$\frac{A^2}{2} \left[(L-0) - \frac{L}{2n\pi} (\sin 2n\pi - \sin 0) \right] = 1$$

$$n = 1, 2, 3, \dots$$

$$\frac{A^2}{2} L = 1$$

$$\Rightarrow A^2 = \frac{2}{L}$$

$$A = \sqrt{\frac{2}{L}}$$

Put in eq (5)

$$\boxed{\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}}$$

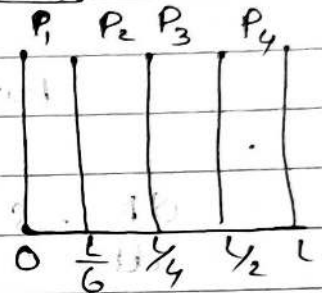
This is wave function of particle in 1-D Box.

Probability Current:

$$s = \frac{dp}{dt} = \text{probability current}$$

To prove:-
$$s = \frac{-i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

The probability of finding the particle in a region is given by



1 mark
$$P = \int \psi^* \psi dx \quad \dots \dots \textcircled{1}$$

Take derivative of $\textcircled{1}$ both sides w.r.t 't'

$$\frac{dP}{dt} = \frac{d}{dt} \int \psi^* \psi dx$$

$$\frac{dP}{dt} = \int \frac{d}{dt} (\psi^* \psi) dx$$

$\psi^* = \text{complex conjugate of } \psi$

$$\frac{dP}{dt} = \int \left(\psi^* \frac{d\psi}{dt} + \frac{d\psi^*}{dt} \psi \right) dx$$

$$\frac{dP}{dt} = \int \psi^* \frac{d\psi}{dt} dx + \int \psi \frac{d\psi^*}{dt} dx \quad \dots \dots \textcircled{2}$$

Now the time-dependent Schrodinger eqn:

$$\left[i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi \right] \rightarrow \text{Remember}$$

$$\int f(x) g(x) = f(x) \int g(x) dx = \int \frac{df(x)}{dx} \int g(x) dx$$

Date: _____

$$\frac{d\psi}{dt} = \frac{1}{i\hbar} \left[-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi \right] \quad (3)$$

Take complex conjugate both sides

$$\frac{d\psi^*}{dt} = \frac{1}{-i\hbar} \left[-\frac{\hbar^2}{2m} \frac{d^2\psi^*}{dx^2} + V\psi^* \right] \quad (4)$$

Put the value (3) & (4) in eq (2)

$$\frac{dP}{dt} = S = \int \psi^* \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi \right) dx + \frac{\psi}{-i\hbar} \left(-\frac{\hbar^2}{2m} \frac{d^2\psi^*}{dx^2} + V\psi^* \right) dx$$

$$= \int \left[\frac{-\hbar}{2mi} \psi^* \frac{d^2\psi}{dx^2} + \frac{\psi^* V \psi}{i\hbar} + \frac{\hbar}{2mi} \frac{\psi^2 d\psi^*}{dx^2} - \frac{1}{i\hbar} \psi V \psi^* \right] dx$$

$$= \frac{-\hbar}{2mi} \int \left(\psi^* \frac{d^2\psi}{dx^2} - \psi \frac{d^2\psi^*}{dx^2} \right) dx$$

$$= \frac{-\hbar}{2mi} \left\{ \left[\psi^* \frac{d\psi}{dx} - \int \frac{d\psi^*}{dx} \frac{d\psi}{dx} dx \right] - \left[\psi \frac{d\psi^*}{dx} - \int \frac{d\psi}{dx} \frac{d\psi^*}{dx} dx \right] \right\}$$

$$= \left[\psi^* \frac{d\psi}{dx} - \int \frac{d\psi^*}{dx} \frac{d\psi}{dx} dx - \psi \frac{d\psi^*}{dx} + \int \frac{d\psi}{dx} \frac{d\psi^*}{dx} dx \right]$$

$$S = \frac{dP}{dt} = \frac{i\hbar}{2m} \left[\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right] \rightarrow \text{Probability Current.}$$

Wave packet

→ wave packet is a localized wave created by superposition of multiple of different frequency.

(1) Phase velocity $V_p = \frac{\omega}{k}$

→ Phase velocity is the velocity at which a single point of wave propagates.

(2) Particle velocity

→ It is a velocity of every particle in the medium oscillates due to wave.

(3) Group velocity $V_g = \frac{d\omega}{dk}$

ω = Angular frequency

k = wave vector

$$k = \frac{2\pi}{\lambda}$$

→ The velocity with which the overall shape or envelope of wave packet propagates.

N.V.S Relation Between Phase velocity and Group velocity

Phase velocity $v_p = \frac{\omega}{k} \Rightarrow \omega = k v_p$

Group velocity $v_g = \frac{d\omega}{dk}$

$$v_g = \frac{d\omega}{dk}$$

$$k = \frac{2\pi}{\lambda}$$

$$v_g = \frac{d}{dk} (k v_p)$$

$$\frac{dk}{d\lambda} = -\frac{2\pi}{\lambda^2}$$

$$v_g = k \frac{dv_p}{dk} + \frac{dk}{dk} v_p$$

$$\frac{d\lambda}{dk} = -\frac{\lambda^2}{2\pi}$$

Put $\frac{d\lambda}{dk} = -\frac{\lambda^2}{2\pi}$ in R.H.S. - (1)

$$v_g = v_p + k^2 \frac{dv_p}{dk}$$

$$v_g = v_p + \frac{2\pi}{\lambda} \cdot \left(-\frac{\lambda^2}{2\pi} \right) \frac{dv_p}{d\lambda}$$

$$v_g = v_p + k \frac{dv_p}{d\lambda} \cdot \frac{d\lambda}{dk} \text{ (1)}$$

$$v_g = v_p - \lambda \frac{dv_p}{d\lambda}$$

Q. mark

Ques. Define (a) phase velocity (b) Group velocity
(c) Particle velocity.

finite Potential value well.

$$V_0 > 0 \quad 0 < x < L$$

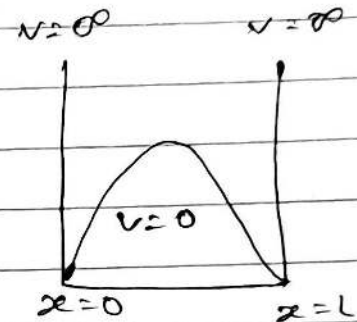
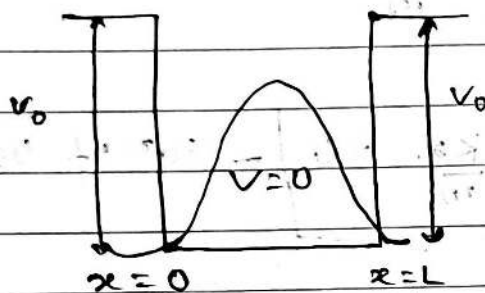
$$V = 0 \quad x \leq 0 \text{ and } x \geq L$$

$$x > 0$$

$$V = V_0 \quad x \leq 0$$

$$V = +\infty \quad x \leq 0$$

$$x \geq 0$$

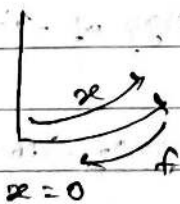


1 marks

Turneling in finite Potential well

Turneling is a quantum mechanical phenomenon where particles has non-zero probability of penetrating a potential energy barrier, if its total energy is less than the ~~use~~ barrier energy.

Harmonic Oscillator



Restoring force $f \propto x$

$$f = -Kx$$

K = force constant

By Newton's law

$$F = ma$$

$$ma = -kx$$

$$m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\boxed{\frac{d^2x}{dt^2} + \frac{k}{m}x = 0} \text{ - Eqn of Oscillator}$$

$$\text{Put } \omega^2 = \frac{k}{m} \Rightarrow \boxed{k = m\omega^2}$$

$$\text{Now eqn becomes } \boxed{\frac{d^2x}{dt^2} + \omega^2x = 0}$$

marks Energy of Quantum Harmonic Oscillator

$$\boxed{E_n = \left(n + \frac{1}{2}\right) \hbar \omega}$$

$n = 0, 1, 2, 3, \dots$

$$n=0: E_0 = \frac{1}{2} \hbar \omega = \text{ground state energy}$$

$$n=1: E_1 = \frac{3}{2} \hbar \omega = \text{Energy of first excited state}$$

$$n=2: E_2 = \frac{5}{2} \hbar \omega = \text{Energy of second excited state}$$

UNIT-IV B

Classical free Electron Theory (Drude Theory)

→ when we apply electric field between two ends of metal, then the force acting on e^- is given by

$$F = eE \quad \text{--- (1)}$$

according to Newton's law

$$F = ma \quad \text{--- (2)}$$

$$ma = eE$$

$$a = \frac{eE}{m} \quad \text{--- (3)}$$

In the presence of electric field drift velocity of e^- is

$$V_d = a\tau \quad \text{--- (4)}$$

V_d = drift velocity

a = acceleration

using eq (3)

τ = collision time

$$V_d = \frac{eE\tau}{m} \quad \text{--- (5)}$$

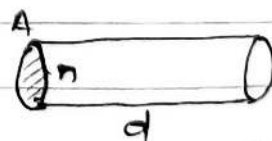
Hence, collision time τ is given by

$$\tau = \frac{\lambda}{\bar{v}}$$

λ = collision length

$\bar{v}$ = mean square speed = $\sqrt{\frac{3kT}{m}}$

$$V_d = \frac{eE}{m} \cdot \frac{\lambda}{\bar{v}} \quad \text{--- (6)}$$



→ for a metal having length 'd' and area of cross section 'A' total number of charge on the metal is $= neAd$

Also current I is

$$I = \frac{\text{charge}}{\text{time}} = \frac{neAd}{t}$$

$$I = neAv_d \quad \text{--- (7)} \quad \therefore v_d = \frac{d}{t} = \text{drift velocity}$$

Now the current density J is between two ends of given by

$$J = \sigma E, \quad \sigma = \text{conductivity}$$

$$\frac{I}{A} = \sigma E$$

using eq (7)

$$\frac{neAv_d}{A} = \sigma E$$

$$nev_d = \sigma E$$

using eq -- (6)

$$ne \frac{e\lambda}{m} \frac{1}{v} = \sigma E$$

$$\boxed{\frac{ne^2\lambda}{m v} = \sigma} \quad \text{--- (8)}$$

$$\text{Put } v = \sqrt{\frac{3KT}{m}}$$

$$\frac{ne^2\lambda}{m} \frac{\sqrt{m}}{\sqrt{3KT}} = \sigma$$

$$\frac{ne^2\lambda}{\sqrt{m^3 3KT}} = \sigma$$

$$\sigma = \frac{ne^2 \lambda}{\sqrt{3mkT}} \quad \text{--- (8)}$$

conductivity.

$$\text{Resistivity } \rho = \frac{1}{\sigma}$$

$$\rho = \frac{m\bar{v}}{ne^2 \lambda} \quad \text{--- (9)}$$

$$\rho = \frac{\sqrt{3mkT}}{ne^2 \lambda} \quad \text{--- (9)}$$

$$\text{mobility } \mu = \frac{V_d}{E} \quad \text{--- (10)}$$

$$V_d = \frac{eE\tau}{m} \quad \text{--- using eq (5)}$$

$$\text{Put in eq (10)}$$

$$\mu = \frac{eE\tau}{m} \cdot \frac{1}{E}$$

$$\mu = \frac{e\tau}{m}$$

$$\tau = \frac{m\mu}{e} \quad \text{--- (11)}$$

Since,

$$\sigma = \frac{ne^2}{m} \frac{\lambda}{\bar{v}} = \frac{ne^2 \tau}{m}$$

using eq (11)

$$\sigma = \frac{ne^2}{m} \cdot \frac{m\mu}{e}$$

$$\sigma = ne\mu$$

This is the relation between conductivity and mobility.

when a potential difference V is apply

$$R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$$

$$E = \frac{V}{l}$$

Also,

$$J = \sigma E$$

$$\frac{I}{A} = \frac{E}{\rho} \quad \text{--- in previous}$$

$$\frac{I}{A} = \frac{V}{l} \cdot \frac{l}{RA}$$

$$\boxed{V = IR} \rightarrow \text{Ohm's law}$$

Quantum free electron theory (Same as particle in 1-Dimensional Box)

→ Electrons are moving freely inside the box of length 'L'.

$$\Psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad \hbar = \frac{h}{2\pi}$$

fermi Energy

→ The energy ~~upto~~ below which energy states are filled with electrons and above which energy states are empty at 0K is called fermi energy or fermi level.

Ques: Derive ~~and~~ an expression for density of state and fermi-dirac distribution. (Previous year)

Ans: $D(E)$ Density of states defined as number of electronic states per unit energy range.

$$\text{Density of states } D(E) = \frac{2dn}{dE} \quad \text{--- (1)}$$

Here, 2 is written for two spins of e^-

Energy of e^- in n^{th} state is

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{--- (2)}$$

Take derivative both sides

$$dE = \frac{2n\pi^2\hbar^2}{2ml^2} dn$$

$$dE = \frac{n\pi^2\hbar^2}{ml^2} dn$$

$$\boxed{\frac{dn}{dE} = \frac{ml^2}{n\pi^2\hbar^2}}$$

Put this value in eq (2)

$$D(E) = \frac{2ml^2}{n\pi^2\hbar^2} \quad (3)$$

Using (2)

$$n^2 = \frac{2ml^2 E}{\pi^2\hbar^2}$$

$$n = \left(\frac{2ml^2}{\pi^2\hbar^2} \right)^{1/2} E^{1/2}$$

Put in (3)

$$D(E) = \frac{2ml^2}{\pi^2\hbar^2} \left(\frac{\pi^2\hbar^2}{2ml^2} \right)^{1/2} \cdot \frac{1}{E^{1/2}}$$

$$= \frac{2ml^2}{\pi^2\hbar^2} \cdot \frac{\pi\hbar}{l} \left(\frac{1}{2m} \right)^{1/2} \frac{1}{E^{1/2}}$$

$$= \frac{(2m)L}{\pi \hbar} (2m)^{-1/2} \cdot \frac{1}{e^{1/2}}$$

$$D(E) = \frac{L}{\pi \hbar} \frac{(2m)^{1/2}}{e^{1/2}}$$

$$D(E) \propto E^{-1/2}$$

V.U.I Fermi-Dirac Distribution

→ It is given by —

$$F(E) = \frac{1}{1 + e^{(E - E_f)/k_B T}}$$

E_f = Fermi Energy

k_B = Boltzmann constant

T = Temperature

(i) at 0 K, $E < E_f \Rightarrow E - E_f = -ve$

$$F(E) = \frac{1}{1 + e^{-ve/0}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + e^{\frac{1}{\infty}}}$$

$$e^{\infty} = \infty$$

$$F(E) = \frac{1}{1 + \frac{1}{\infty}} = \frac{1}{1 + 0} = 1$$

(ii) at 0 K, $E > E_f$

$$F(E) = \frac{1}{1 + e^{+ve/0}}$$

$$= \frac{1}{1+e^{\infty}} = \frac{1}{1+\infty} = \frac{1}{\infty}$$

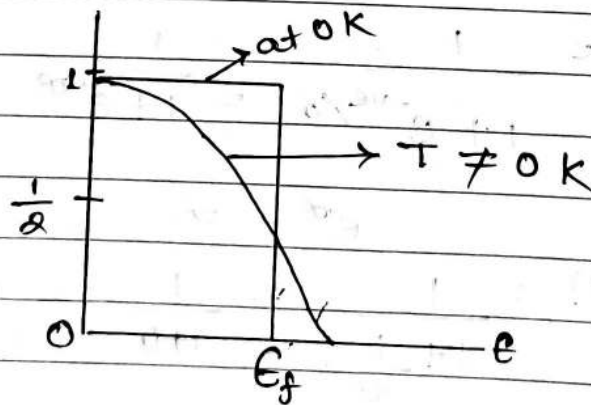
$$F(E) = 0$$

(iii) at 0 K, $E = E_f$ $\Rightarrow E - E_f = 0$

$$F(E) = \frac{1}{1+e^0} = \frac{1}{1+1} = \frac{1}{2}$$

(iv) $E > E_f$, $T \neq 0$ K, $E - E_f = +ve$

$$F(E) = \frac{1}{1+e^{+ve/T}} \neq 0$$



failure or limitation of Quantum Free

- It is unable to distinguish between insulator, semiconductor and conductor.
- It is not able to explain why some solids are conductors.
- It doesn't explain metallic properties of crystals.
- It is unable to explain positive value hall co-efficient.

Thermionic Emission and Richardson's Equation

- let us suppose that motion of electron is along X-axis.
- The phenomena of emission of electrons from the surface of metal, when the metal is heated is called thermionic emission.
- It will occur when the K.E of electron along X-axis.

$$\frac{1}{2} m v^2 \geq E_f + \phi \quad \phi = \text{work function}$$

$$\text{Minimum energy } \frac{1}{2} m v_x^2 = E_f + \phi$$

$$\boxed{v_x^2 = \frac{2(E_f + \phi)}{m}} \quad \text{minimum} \quad \dots (1)$$

The density of states of electrons per unit volume in the momentum range " p to $p+dp$ " is given by

$$N = \frac{8\pi p^2 dp}{h^3 [1 + e^{(\epsilon - \epsilon_f)/k_B T}]} \quad \text{--- (2)}$$

p = linear momentum

$p = mv$

$dp = m dv$

Put these value in (2)

$$N = \frac{8\pi m^2 v^2 m dv}{h^3 [1 + e^{(\epsilon - \epsilon_f)/k_B T}]}$$

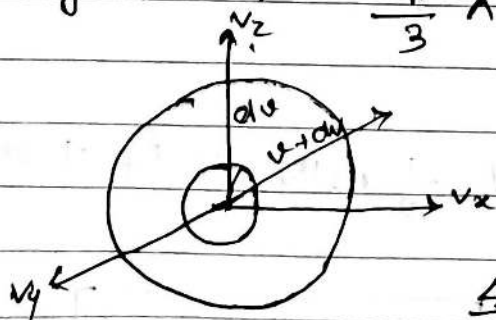
$$N = \frac{8\pi m^3 v^2 dv}{h^3 [1 + e^{(\epsilon - \epsilon_f)/k_B T}]} \quad \text{--- (3)}$$

If v_x , v_y and v_z are x , y and z components of velocity then the space velocity of electrons in 3D in the range v_x to $v_x + dv_x$

v_y to $v_y + dv_y$

v_z to $v_z + dv_z$

is given by $= \frac{4}{3} \pi [(v+dv)^3 - v^3]$



$$\frac{4}{3} \pi v^3$$

$$\frac{4}{3} \pi (v+dv)^3 - \frac{4}{3} \pi v^3$$

$$\begin{aligned}
 dv_x dv_y dv_z &= \frac{4}{3} \pi [(v+dv)^3 - v^3] \\
 &= \frac{4}{3} \pi [v^3 + dv^3 + 3dv(v+dv) - v^3] \\
 &= \frac{4}{3} \pi [(dv^3) + 3v^2 dv + 3v(dv^2)]
 \end{aligned}$$

If dv is very small $\Rightarrow dv^2$ and dv^3 are very very small.

So, we can neglect dv^2 and dv^3

$$\begin{aligned}
 dv_x dv_y dv_z &= \frac{4}{3} \pi \times 3v^2 dv \\
 &= 4\pi v^2 dv
 \end{aligned}$$

$$v^2 dv = \frac{dv_x dv_y dv_z}{4\pi}$$

put this value in eq (3)

$$N = \frac{2\pi m^3}{h^3 \times 4\pi} \frac{dv_x dv_y dv_z}{[1 + e^{(E-E_f)/k_B T}]}$$

$$N = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{[1 + e^{(E-E_f)/k_B T}]} \quad \text{--- (4)}$$

When we heat the metal surface, then E will become very large i.e., $E - E_f$ and hence $(E - E_f)/k_B T$ will become very large.

Hence, we can neglect 1 in comparison to $e^{(E-E_f)/k_B T}$

$$N = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{e^{(E-E_f)/k_B T}} = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{e^{(E/k_B T - E_f/k_B T)}}$$

$$N = \frac{2m^3}{h^3} e^{-E/k_B T} e^{E_f/k_B T} dx dy dz$$

$$\text{Put } A = \frac{2m^3}{h^3} e^{E_f/k_B T}$$

$$N = A(e)^{-E/k_B T} dx dy dz$$

The current density of electrons emitted along x-axis is given by $J_x = Ne v_x$

$$J_x = A e^{-E/k_B T} e v_x$$

$$E_x = \frac{1}{2} m v_x^2$$

$$E_y = \frac{1}{2} m v_y^2$$

$$E_z = \frac{1}{2} m v_z^2$$

$$J_x = A e^{-\frac{1}{2} \frac{m v_x^2}{k_B T} - \frac{1}{2} \frac{m v_y^2}{k_B T} - \frac{1}{2} \frac{m v_z^2}{k_B T}}$$

$$J_x = A \int v_x e^{-\frac{1}{2} \frac{m v_x^2}{k_B T}} dx \int e^{-\frac{1}{2} \frac{m v_y^2}{k_B T}} dy \int e^{-\frac{1}{2} \frac{m v_z^2}{k_B T}}$$

Imarks

$$J_x = A T^2 e^{-\Phi/k_B T} \rightarrow \text{Richardson's Equation}$$

$\therefore J_x =$ Current density.

UNIT-IV A

Date : _____

Crystal Structure

Lattice parameter of a unit cell.

- ① distance in all direction. $x, y, z \rightarrow$ axis
- ② angle α, β, γ

Bravais Lattice

→ The lattice where in which every lattice point has identical surrounding and obeys symmetric operation is called Bravais Lattice.

There are five Bravais lattice in 2-D and there are fourteen Bravais lattice in 3-D.

2-D

$a = b, \gamma = 90^\circ$ square

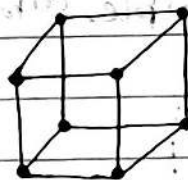
$a = b, \gamma = 120^\circ$ 60° rhombus

$a \neq b, \gamma \neq 90^\circ$ parallelogram

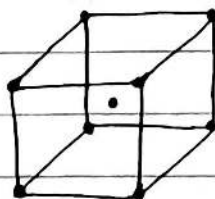
$a \neq b, \gamma = 90^\circ$ rectangle

3-D

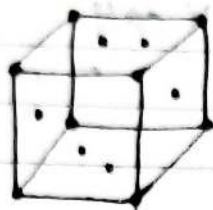
① simple cubic (SCC)



② Body Centred cubic (BCC)



⑧ face centred cube (fcc)



No of atoms per unit cell

$$N = \frac{N_c}{8} + \frac{N_b}{1} + \frac{N_f}{2}$$

$$① \text{ SSC } N = \frac{8}{8} + 0 + 0 = 1$$

$$② \text{ BCC } N = \frac{8}{8} + \frac{1}{1} + 0 = 2$$

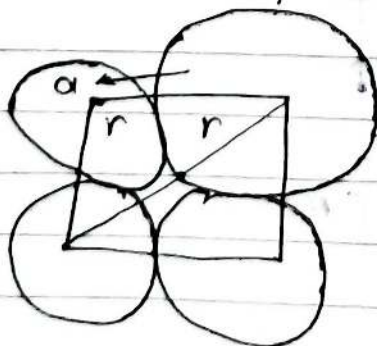
$$③ \text{ FCC } N = \frac{8}{8} + 0 + \frac{6}{2} = 1 + 3 = 4$$

Ques. Define the term bravais's lattice, crystal,

Atomic Radius :- It is defined as the distance between two nearest neighbouring atoms.

Atomic Radius

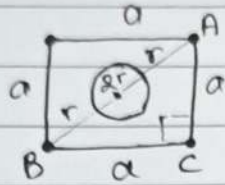
① Atomic Radius of simple cubic



$$2r = a$$

$$r_{sc} = \frac{a}{2}$$

⑤ fcc



$$(AB)^2 = BC^2 + AC^2$$

$$(4r)^2 = a^2 + a^2$$

$$4r = \sqrt{2a^2}$$

$$r = \frac{\sqrt{2}a}{4}$$

$$r = \frac{\sqrt{2}}{4} \times \frac{\sqrt{2}}{\sqrt{2}} a = \frac{2a}{2\sqrt{2}} = \frac{a}{\sqrt{2}}$$

The no. of neighbouring atoms that surround every atom in the lattice is called co-ordination number.